

EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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Summary: Data arising from life-testing and reliability studies are often left truncated and right censored. Some of the most commonly used distributions to model lifetime data are the lognormal, Weibull, gamma and exponential distributions. Here, the EM algorithm is used to estimate the parameters of the lognormal, Weibull and gamma models based on left truncated and right censored data. The parsimonious model that includes the lognormal, Weibull, exponential and gamma distributions as special cases is the generalized gamma distribution. The EM algorithm steps for the generalized gamma distribution are also derived based on left truncated and right censored data. The asymptotic variance-covariance matrices of the MLEs are derived by using the missing information principle (Louis, 1982), and then the asymptotic confidence intervals for the parameters are obtained. The Newton-Raphson method is also applied to obtain the MLEs for the lognormal, Weibull and gamma distributions, for comparison purpose. The methods of inference are compared through extensive Monte Carlo simulation studies, and some numerical examples are given to illustrate all the methods of inference developed here. A model discrimination problem is addressed using the information-based criteria.

1. Introduction

Truncation and censoring are quite common in lifetime data; interested readers may refer to Balakrishnan and Cohen (1991), Cohen (1991) and Meeker and Escobar (1998) for details. In this paper,

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we consider left truncated and right censored lifetime data. Recently, Hong, Meeker and McCalley (2009) analyzed left truncated and right censored data pertaining to the lifetimes of power transformers in an electrical industry in the United States. In their work, the Weibull distribution turned out to be the most appropriate model. However, for modelling lifetime data, some other distributions which are commonly used are the lognormal, gamma and exponential etc; see Johnson, Kotz and Balakrishnan (1994), Crow and Shimizu (1988), Bowman and Shenton (1988) and the references therein.

The Expectation Maximization (EM) algorithm is a powerful tool for analyzing incomplete data; see McLachlan and Krishnan (2008). In this paper, the detailed steps of the EM algorithm are derived for estimating the parameters of the lognormal, Weibull and gamma distributions based on left truncated and right censored data. Under the EM framework, the asymptotic variance-covariance matrices of the maximum likelihood estimates (MLEs) are obtained by using the missing information principle (Louis, 1982). Then, by using the asymptotic normality of the MLEs, the asymptotic confidence intervals for the model parameters are constructed. For comparison purpose, the Newton-Raphson method is also used to obtain the MLEs, and then the asymptotic confidence intervals corresponding to the observed likelihood are also constructed for the parameters. Through extensive Monte Carlo simulations, for all the models, the methods of estimation and the asymptotic confidence intervals, in terms of the coverage probability, are compared.

The parsimonious model that includes the exponential, lognormal, Weibull and gamma distributions as special cases is the generalized gamma distribution; see Johnson et al. (1994). The EM algorithm steps are also derived for the generalized gamma distribution, based on left truncated and right censored lifetime data. For a given data, it might be of interest to know which of these models provides a better fit. To address this question, a model discrimination analysis is carried out by using the information criteria. Also an application of this estimation technique to prediction is discussed.

The rest of this paper is organized as follows. In Section 2, the form of the data and the observed likelihood are presented. In Section 3, the EM algorithm in general and the construction of asymptotic variance-covariance matrix of the MLEs using the missing information principle are discussed. In this section, the construction of parametric bootstrap confidence interval and an application to prediction are also discussed. Sections 4, 5, 6 and 7 present the specific derivations for the inference for the lognormal, Weibull, gamma and generalized gamma distributions, respectively. Section 8 presents the simulation results, while Section 9 presents illustrative examples for all the methods of inference developed here for all the distributions under consideration. In Section 10, the model discrimination analysis is carried out. Finally, some concluding remarks and possible future works are mentioned in Section 11.

2. Form of the data and the observed likelihood

The Monte Carlo simulations are set up here to mimic the dataset used by Hong et al. (2009). The data can be considered to be lifetime data of power transformers in an electrical industry, over a particular interval of time. The failure time of a machine is observed only if it fails after 1980, as detailed record keeping on the lifetime of machines was started only in that year. Information on the lifetime of machines installed after 1980 are available, while for machines installed before 1980,

lifetimes are available only if they failed after 1980, but no information is available on machines that failed before 1980. This results in the data being left truncated. Also, the lifetime of the machines are followed only until 2008, which incorporates right censoring into the data. The machines which failed before 2008 were completely observed, while the machines that did not fail until 2008 resulted in right censored observations.

Here, we describe the notation used throughout this paper. Let Y denote the lifetime variable, C denote the censoring time variable, δ_i denote the censoring indicator, i.e., δ_i is 0 if the i -th observation is censored and 1 if the observation is not censored, and τ_i^L denote the left truncation time. To be precise, for a machine installed before 1980, τ_i^L is the time between the year of installation and the truncation point of 1980. Let v_i denote the truncation indicator, i.e., v_i is 0 if an observation is truncated and 1 if it is not truncated. Further, let S_1 and S_2 be two index sets, where S_1 is the set of machines installed after 1980 and S_2 is the set of machines installed on or before 1980.

When the parameter vector of interest is denoted by θ , the likelihood function for the left truncated and right censored data is given by

$$L(\theta) = \prod_{i \in S_1} \{f_Y(y_i)\}^{\delta_i} \{1 - F_Y(y_i)\}^{1-\delta_i} \times \prod_{i \in S_2} \left\{ \frac{f_Y(y_i)}{1 - F_Y(\tau_i^L)} \right\}^{\delta_i} \left\{ \frac{1 - F_Y(y_i)}{1 - F_Y(\tau_i^L)} \right\}^{1-\delta_i}.$$

In the following sections while developing the steps of the EM-algorithm for different lifetime distributions, in some cases we use log-transformed variable to make the subsequent derivations simpler. To be precise, in Sections 4 and 5, for lognormal and Weibull distributions, we log-transform the data to obtain normal and extreme value models, respectively. In Sections 6 and 7, for gamma and generalized gamma distributions, the original scale itself is kept as they are convenient to work with. Consequently, in Sections 4 and 5, the above symbols will be for the log-transformed variables while for Sections 6 and 7, they will correspond to the original scale variables.

3. The EM algorithm

The Expectation Maximisation (EM) algorithm is a very powerful and useful tool for analyzing incomplete data; see McLachlan and Krishnan (2008) for an elaborate discussion on this method. The algorithm consists of two steps - the Expectation step (E-step) and the Maximization step (M-step). First of all, the complete data likelihood for the given problem is formed. Then, in the E-step, the conditional expectation of the complete data log-likelihood is obtained, given the observed incomplete data and the current value of the parameter, real- or vector-valued. This expected log-likelihood is essentially a function of the parameter involved and the current value of the parameter under which the expectation has been calculated. Thus, if the underlying lifetime variable is T and the observed variable is Y , then in the E-step, our objective is to obtain

$$Q(\theta, \theta^{(k)}) = E_{\theta^{(k)}} [\log L_c(\mathbf{t}; \theta) | \mathbf{y}],$$

where $L_c(\mathbf{t}; \theta)$ is the complete data likelihood, θ is the parameter vector of interest and $\theta^{(k)}$ is the value of the parameter vector at the k -th step of the algorithm (that is, the current value of the parameter).

In the M-step, this expected complete data log-likelihood $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)})$ is maximized with respect to $\boldsymbol{\theta}$ over the parameter space Θ to obtain the improved estimate of the parameter as

$$\boldsymbol{\theta}^{(k+1)} = \arg \max_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(k)}).$$

Depending on the form of the Q -function, the parameter estimate could be in closed-form, or we may have to employ some numerical methods.

The E- and M-steps are then iterated till convergence to the desired level of accuracy. This algorithm is known to have many desirable and advantageous properties over the direct method for obtaining the MLEs in the case of incomplete data; one may consult McLachlan and Krishnan (2008) for pertinent details.

3.1. Asymptotic variance-covariance matrix of the MLEs

The asymptotic variance-covariance matrix of the MLEs under the EM framework is not directly obtained. Louis (1982) developed a technique to be used under the EM framework for obtaining the asymptotic variance-covariance matrix of the MLEs, based on the missing information principle. According to this principle,

$$\text{Observed information} = \text{Complete information} - \text{Missing information}.$$

Let C and δ denote the censoring time and censoring indicator, respectively; i.e., $\delta = 0$ if a unit is censored, and it is 1 if the unit is not censored. Further, let $I_T(\boldsymbol{\theta})$, $I_Y(\boldsymbol{\theta})$ and $I_{C|Y}(\boldsymbol{\theta})$ denote the complete information matrix, the observed information matrix and the missing information matrix, respectively. The complete information matrix is given by

$$I_T(\boldsymbol{\theta}) = -E \left[\frac{\partial^2}{\partial \boldsymbol{\theta}^2} \log L_c(\mathbf{t}; \boldsymbol{\theta}) \right].$$

The Fisher information matrix for the i -th observation which is censored is given by

$$I_{C|Y}^{(i)}(\boldsymbol{\theta}) = -E \left[\frac{\partial^2}{\partial \boldsymbol{\theta}^2} \log f_{C_i}(c_i | C_i > y_i, \boldsymbol{\theta}) \right].$$

The expected missing information can thus be obtained as

$$I_{C|Y}(\boldsymbol{\theta}) = \sum_{i: \delta_i=0} I_{C|Y}^{(i)}(\boldsymbol{\theta}).$$

Therefore, by the missing information principle, the observed information matrix is given by

$$I_Y(\boldsymbol{\theta}) = I_T(\boldsymbol{\theta}) - I_{C|Y}(\boldsymbol{\theta}).$$

The asymptotic variance-covariance matrix of the MLE of $\boldsymbol{\theta}$, under the EM framework, can be obtained by inverting the observed information matrix $I_Y(\boldsymbol{\theta})$ at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$.

3.2. Confidence intervals

After obtaining the MLEs of the parameters and the asymptotic variances and covariance of the MLEs, one can construct the asymptotic confidence intervals for the parameters by using the asymptotic normality of the MLEs. For obtaining the asymptotic variance-covariance matrix, one can use the missing information principle as described above, or simply use the observed information matrix directly from the second derivatives of the log-likelihood function.

It is also possible to construct parametric bootstrap confidence intervals for the parameters as follows. First, from the given data, the MLE $\hat{\theta}$ of θ is obtained. Then, using $\hat{\theta}$ as the true value of the parameter θ , a sample of size n in the same sampling framework with left truncation and right censoring is produced. This process is repeated for B Monte Carlo runs and the MLEs are determined for each simulated sample. In the next step, based on these B estimates, the bootstrap bias and variance of the estimates of the parameters are obtained. Finally, the $100(1 - \alpha)\%$ parametric bootstrap confidence interval for μ (a component of θ) is then obtained as

$$\text{LCL: } \hat{\mu} - b_{\mu} - z_{\alpha/2}\sqrt{v_{\mu}}, \quad \text{UCL: } \hat{\mu} - b_{\mu} + z_{\alpha/2}\sqrt{v_{\mu}},$$

where b_{μ} and v_{μ} are, respectively, the bootstrap bias and variance of the parameter μ and z_{α} is the upper α -percentage point of the standard normal distribution.

It would also be possible to construct nonparametric bootstrap confidence intervals, but with both censoring and truncation present in the data, the nonparametric bootstrap method will not be as efficient as a parametric bootstrap method unless the sample size is very large. A similar observation has been made by Balakrishnan, Kundu, Ng and Kannan (2007) in the context of analysis of censored data from step-stress reliability experiments.

3.3. An application to prediction

With the estimated parameters, one can obtain the probability of a censored unit working till a future year, given that it has worked till Y_{cen} (the right censoring point). Suppose a unit is installed in the year Y_{ins} , before 1980, i.e., the unit is left truncated. Then the probability that this unit will be working till a future year Y_{fut} , given that it is right censored at Y_{cen} , will be given by

$$\pi = \frac{S^*(\log(Y_{fut} - Y_{ins}))}{S^*(\log(Y_{cen} - Y_{ins}))},$$

where $S^*(\cdot)$ is the survival function of the left truncated log-transformed random variable. Clearly, the above probability reduces to

$$\pi = \frac{S(\log(Y_{fut} - Y_{ins}))}{S(\log(Y_{cen} - Y_{ins}))} = g(\theta),$$

where $S(\cdot)$ is the survival function of the untruncated log-transformed lifetime variable, and $g(\cdot)$ is a function of θ . Incidentally, this is also the probability of the same event for a unit which is not left truncated. One can obtain an estimate $\hat{\pi}$ by using the MLE $\hat{\theta}$ as $\hat{\pi} = g(\hat{\theta})$.

Using the delta method, and the asymptotic variance-covariance matrix of the MLE $\hat{\theta}$, we can also estimate the variance of the above estimate $\hat{\pi}$. A straightforward application of the delta-method

results in

$$\widehat{\pi} \sim N(\pi, \text{Var}(\widehat{\pi})),$$

where $\text{Var}(\widehat{\pi})$ can be estimated as

$$\widehat{\text{Var}}(\widehat{\pi}) = (\nabla g)' \Sigma (\nabla g) \Big|_{\theta = \widehat{\theta}}, \quad (3.1)$$

with ∇g denoting the vector of partial derivatives of g with respect to the parameters and Σ denoting the asymptotic variance-covariance matrix of the MLE $\widehat{\theta}$. Examples of this technique are given in the illustrative examples in Section 9.

4. Inference for the lognormal distribution

The pdf of a lognormal random variable X is given by (see Johnson et al., 1994)

$$f_X(x) = \frac{1}{\sigma x \sqrt{2\pi}} \exp \left\{ -\frac{(\log x - \mu)^2}{2\sigma^2} \right\}, \quad x > 0,$$

where μ and σ are the scale and shape parameters, respectively. Then, the variable $Y = \log X$ is normally distributed with density

$$f_Y(y) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}, \quad -\infty < y < \infty,$$

where μ and σ are the location and scale parameters, respectively, of the log-variable Y . Let $\phi(\cdot)$ and $\Phi(\cdot)$ denote the probability density function and cumulative distribution function of the standard normal distribution, respectively.

4.1. Estimation of parameters

The first step to employ the EM algorithm is to obtain the complete data likelihood. Let T denote the log-transformed lifetime of a machine, and θ denote the parameter vector $(\mu, \sigma)'$. With all the other notation remaining as before, had there been no censoring, the complete data likelihood would be

$$L_c(\mathbf{t}; \theta) = \prod_{s_1} \left\{ \frac{1}{\sigma} \phi \left(\frac{t_i - \mu}{\sigma} \right) \right\} \times \prod_{s_2} \left\{ \frac{\frac{1}{\sigma} \phi \left(\frac{t_i - \mu}{\sigma} \right)}{1 - \Phi \left(\frac{\tau_i^L - \mu}{\sigma} \right)} \right\}.$$

The log-likelihood function, using the truncation indicator v_i , is given by

$$\log L_c(\mathbf{t}; \theta) = -n \log \sigma - \frac{1}{2\sigma^2} \sum_{i=1}^n (t_i^2 - 2\mu t_i + \mu^2) - \sum_{i=1}^n (1 - v_i) \log \left\{ 1 - \Phi \left(\frac{\tau_i^L - \mu}{\sigma} \right) \right\}. \quad (4.1)$$

Let us denote the vector of censoring indicators by $\delta = (\delta_1, \delta_2, \dots, \delta_n)'$, and the observed data vector by $\mathbf{y} = (y_1, y_2, \dots, y_n)'$, where $y_i = \min(t_i, c_i)$, i.e., the minimum of the two variables t_i and c_i , as the data are right censored.

The required conditional expectations can be obtained, using the results of Cohen (1991), as follows:

$$E_{\theta^{(k)}}[t_i | t_i > y_i] = \mu^{(k)} + \sigma^{(k)} q_i^{(k)},$$

$$E_{\theta^{(k)}}[t_i^2 | t_i > y_i] = \sigma^{(k)2} (1 + \xi_i^{(k)} q_i^{(k)}) + 2\sigma^{(k)} \mu^{(k)} q_i^{(k)} + \mu^{(k)2},$$

where

$$\xi_i^{(k)} = \frac{y_i - \mu^{(k)}}{\sigma^{(k)}},$$

$$q_i^{(k)} = \frac{\phi(\xi_i^{(k)})}{1 - \Phi(\xi_i^{(k)})},$$

with $\theta^{(k)} = (\mu^{(k)}, \sigma^{(k)})$ being the parameter value at the k -th iteration step. Substituting these conditional expectations in Eq. (4.1), the expected complete data likelihood is obtained to be

$$\begin{aligned} Q(\theta, \theta^{(k)}) = & -n \log \sigma - \sum_{i=1}^n (1 - v_i) \log \left\{ 1 - \Phi \left(\frac{\tau_i^L - \mu}{\sigma} \right) \right\} \\ & - \frac{1}{2\sigma^2} \left[\sum_{i:\delta_i=1} y_i^2 + \sum_{i:\delta_i=0} \{ \sigma^{(k)2} (1 + \xi_i^{(k)} q_i^{(k)}) + 2\sigma^{(k)} \mu^{(k)} q_i^{(k)} + \mu^{(k)2} \} \right] \\ & + \frac{\mu}{\sigma^2} \left[\sum_{i:\delta_i=1} y_i + \sum_{i:\delta_i=0} \{ \mu^{(k)} + \sigma^{(k)} q_i^{(k)} \} \right] - \frac{n\mu^2}{2\sigma^2}. \end{aligned} \quad (4.2)$$

In the M-step, (4.2) is maximized with respect to the parameters to obtain the MLEs. In the work of Balakrishnan and Mitra (2011), the M-step of the EM algorithm has been carried out in two different ways as follows.

Method 1. The first-order derivatives of the function $Q(\theta, \theta^{(k)})$ with respect to μ and σ involve the hazard function of the normal distribution along with polynomial components in μ and σ , thus making the solution for μ and σ to be quite involved. The equations obtained by equating the first-order derivatives of $Q(\theta, \theta^{(k)})$ with respect to μ and σ , respectively, to zero are as follows:

$$n\mu - C_1 + \sigma \sum_{i=1}^n (1 - v_i) h_i(\mu, \sigma) = 0, \quad (4.3)$$

$$n\sigma^2 + \sigma \sum_{i=1}^n (1 - v_i) (\tau_i^L - \mu) h_i(\mu, \sigma) - C_2 + 2\mu C_1 - n\mu^2 = 0, \quad (4.4)$$

where

$$h_i(\mu, \sigma) = \frac{\phi(\frac{\tau_i^L - \mu}{\sigma})}{1 - \Phi(\frac{\tau_i^L - \mu}{\sigma})},$$

$$C_1 = \sum_{i:\delta_i=1} y_i + \sum_{i:\delta_i=0} \{ \mu^{(k)} + \sigma^{(k)} q_i^{(k)} \},$$

$$C_2 = \sum_{i:\delta_i=1} y_i^2 + \sum_{i:\delta_i=0} \{ \sigma^{(k)^2} (1 + \xi_i^{(k)} q_i^{(k)}) + 2\sigma^{(k)} \mu^{(k)} q_i^{(k)} + \mu^{(k)^2} \}.$$

It is evident that explicit expressions cannot be obtained by directly solving Eqs. (4.3) and (4.4). However, an approximation can be made by considering the Taylor series expansion of the hazard function $h_i(\mu, \sigma)$. We consider a Taylor expansion of $h_i(\mu, \sigma)$ with respect to μ for fixed σ to obtain an approximate value of the MLE of μ . In the subsequent step, we consider a Taylor expansion of $h_i(\mu, \sigma)$ with respect to σ for fixed μ to obtain an approximate value for the MLE of σ . This process is then continued iteratively until convergence to obtain the MLEs of μ and σ . Note that these are only approximations to the MLEs of μ and σ .

Thus, carrying out the simplification, we obtain the $(k+1)$ -th step estimates of μ and σ as

$$\mu^{(k+1)} \approx \frac{\frac{1}{n} \left[C_1 - \sigma^{(k)} \sum_{i=1}^n (1 - v_i) \left\{ h_i(\mu^{(k)}, \sigma^{(k)}) - \mu^{(k)} h_{\mu_i}(\mu^{(k)}, \sigma^{(k)}) \right\} \right]}{1 + \frac{\sigma^{(k)}}{n} \sum_{i=1}^n (1 - v_i) h_{\mu_i}(\mu^{(k)}, \sigma^{(k)})} \quad (4.5)$$

and

$$\sigma^{(k+1)} \approx \frac{-B + \sqrt{B^2 - 4AK}}{2A}, \quad (4.6)$$

where

$$A = n + \sum_{i=1}^n (1 - v_i) (\tau_i^L - \mu^{(k)}) h_{\sigma_i}(\mu^{(k)}, \sigma^{(k)}),$$

$$B = \sum_{i=1}^n (1 - v_i) (\tau_i^L - \mu^{(k)}) \{ h_i(\mu^{(k)}, \sigma^{(k)}) - \sigma^{(k)} h_{\sigma_i}(\mu^{(k)}, \sigma^{(k)}) \},$$

$$K = -C_2 + 2\mu^{(k)} C_1 - n\{\mu^{(k)}\}^2,$$

h_μ and h_σ being the partial derivatives of the hazard function $h_i(\mu, \sigma)$ with respect to μ and σ , respectively. It should be noted that while solving the quadratic equation for σ , we obtain two roots of which the larger one is as presented in Eqn. (4.6), while the smaller one is inadmissible usually and is therefore discarded. However, in some situations, both roots turn out to be positive; we chose the larger root in (4.6) simply because it is used in an iterative process until the occurrence of convergence to the desired tolerance level.

Method 2. An alternative method used in Balakrishnan and Mitra (2011) for the M-step to maximize $Q(\theta, \theta^{(k)})$ is the EM gradient algorithm due to Lange (1995). In this method, the function $Q(\theta, \theta^{(k)})$ is maximized in the M-step by a one-step Newton-Raphson (NR) method. This algorithm is closely related to the original EM algorithm, and qualifies as a special case of the generalized EM algorithm (Dempster, Laird and Rubin, 1977). The properties of this algorithm are almost the same as those of the EM algorithm. However, unlike the EM algorithm, the EM gradient algorithm depends on the choice of the initial values of the parameters. For details on these methods, see Balakrishnan and Mitra (2011, 2014).

For comparative purpose, we also obtain the MLEs of the parameters by the NR method which is a common and direct approach for obtaining the MLEs. The NR method has been employed here through a default function of the R software, called “maxNR”. It has been observed in the Monte Carlo simulation study that all the methods give quite close results.

4.2. Asymptotic variances and covariance of the MLEs

The appropriate elements of the complete information matrix $I_T(\theta)$ given in Section 3.1, calculating the necessary expectations over of the lifetime variable over the two index sets S_1 and S_2 , are obtained as follows:

$$-E \left[\frac{\partial^2}{\partial \mu^2} \log L_c(\mathbf{t}; \theta) \right] = \frac{n}{\sigma^2} + \sum_{i=1}^n \frac{1-v_i}{\sigma^2} \frac{\{1-\Phi(\beta_i)\} \beta_i \phi(\beta_i) - \{\phi(\beta_i)\}^2}{\{1-\Phi(\beta_i)\}^2},$$

$$-E \left[\frac{\partial^2}{\partial \sigma^2} \log L_c(\mathbf{t}; \theta) \right] = \frac{2n}{\sigma^2} + \sum_{i=1}^n \frac{1-v_i}{\sigma^2} \{\beta_i h_i (1 + \beta_i^2 - \beta_i h_i)\},$$

$$-E \left[\frac{\partial^2}{\partial \mu \partial \sigma} \log L_c(\mathbf{t}; \theta) \right] = \sum_{i=1}^n \frac{1-v_i}{\sigma^2} h_i (1 + \beta_i^2 - \beta_i h_i),$$

where $\beta_i = \frac{t_i - \mu}{\sigma}$ and $h_i = \frac{\phi(\beta_i)}{1-\Phi(\beta_i)}$.

Now, the conditional distribution required for the calculation of the missing information matrix is given by (see Ng, Chan and Balakrishnan, 2002)

$$f_{C_i|(C_i > y_i)}(c_i | C_i > y_i, \mu, \sigma) = \frac{\frac{1}{\sigma} \phi\left(\frac{c_i - \mu}{\sigma}\right)}{1 - \Phi\left(\frac{y_i - \mu}{\sigma}\right)}, \quad c_i \geq y_i.$$

The logarithm of the above truncated normal pdf is

$$\log f_{C_i|(C_i > y_i)}(c_i | C_i > y_i, \mu, \sigma) = \text{const} - \log \sigma - \frac{(c_i - \mu)^2}{2\sigma^2} - \log \left\{ 1 - \Phi\left(\frac{y_i - \mu}{\sigma}\right) \right\}. \quad (4.7)$$

The derivatives of (4.7) with respect to μ and σ are given by

$$\frac{\partial}{\partial \mu} \log f_{C_i|(C_i > y_i)} = \frac{1}{\sigma} \left[\frac{c_i - \mu}{\sigma} - q_i \right],$$

$$\frac{\partial}{\partial \sigma} \log f_{C_i|(C_i > y_i)} = \frac{1}{\sigma} \left[\left(\frac{c_i - \mu}{\sigma} \right)^2 - (1 + \xi_i q_i) \right],$$

where $\xi_i = \frac{y_i - \mu}{\sigma}$ and $q_i = \frac{\phi(\xi_i)}{1-\Phi(\xi_i)}$. It can be shown that (see Cohen, 1991; Ng et al., 2002)

$$E[(C_i - \mu) | C_i > y_i, \mu, \sigma] = \sigma q_i,$$

$$E[(C_i - \mu)^2 | C_i > y_i, \mu, \sigma] = \sigma^2 [1 + \xi_i q_i],$$

$$E[(C_i - \mu)^3 | C_i > y_i, \mu, \sigma] = \sigma^3 [2 + \xi_i^2 q_i],$$

$$E[(C_i - \mu)^4 | C_i > y_i, \mu, \sigma] = \sigma^4 [3(1 + \xi_i q_i) + \xi_i^3 q_i].$$

Using these expectations, the elements of the Fisher information matrix for the i -th observation which is censored can be obtained as

$$E \left[\left(\frac{\partial}{\partial \mu} \log f_{C_i|(C_i > y_i)} \right)^2 \right] = \frac{1}{\sigma^2} [1 + \xi_i q_i - q_i^2], \quad (4.8)$$

$$E \left[\left(\frac{\partial}{\partial \sigma} \log f_{C_i|(C_i > y_i)} \right)^2 \right] = \frac{1}{\sigma^2} [2 + \xi_i q_i (1 - \xi_i q_i + \xi_i^2)], \quad (4.9)$$

$$E \left[\left(\frac{\partial}{\partial \mu} \log f_{C_i|(C_i > y_i)} \right) \left(\frac{\partial}{\partial \sigma} \log f_{C_i|(C_i > y_i)} \right) \right] = \frac{1}{\sigma^2} [q_i + \xi_i q_i (\xi_i - q_i)]. \quad (4.10)$$

Using Eqs. (4.8) – (4.10) and the formulas given in Section 3.1, the matrix $I_Y(\boldsymbol{\theta})$ can be obtained. Finally, the asymptotic variance-covariance matrix of the MLEs can be obtained by inverting the matrix $I_Y(\boldsymbol{\theta})$ evaluated at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$; see Balakrishnan and Mitra (2014) for details.

5. Inference for the Weibull distribution

Let X be a Weibull random variable with scale parameter α and shape parameter η . The density of X is given by (see Johnson, Kotz and Balakrishnan, 1995)

$$f_X(x) = \left(\frac{\eta}{\alpha} \right) \left(\frac{x}{\alpha} \right)^{\eta-1} \exp \left\{ - \left(\frac{x}{\alpha} \right)^\eta \right\}, \quad x > 0, \alpha > 0, \eta > 0.$$

The log-transformed variable $Y = \log X$ follows an extreme value distribution with density

$$f_Y(y) = \frac{1}{\sigma} \exp \left[\left(\frac{y - \mu}{\sigma} \right) - \exp \left(\frac{y - \mu}{\sigma} \right) \right], \quad -\infty < y < \infty, -\infty < \mu < \infty, \sigma > 0,$$

where $\mu = \log \alpha$ and $\sigma = 1/\eta$ are the location and scale parameters, respectively, of the log-variable. The lifetime variable is assumed to follow a Weibull distribution with the above density, and that we log-transform the data in order to use the extreme value model conveniently for the subsequent derivations.

5.1. Estimation of parameters

Under the extreme value model and with the parameter vector denoted by $\boldsymbol{\theta} = (\mu, \sigma)'$, had there been no censoring, the complete data likelihood would be

$$\begin{aligned} L_c(\mathbf{t}; \boldsymbol{\theta}) &= \prod_{S_1} \left\{ \frac{1}{\sigma} \exp \left[\left(\frac{t_i - \mu}{\sigma} \right) - \exp \left(\frac{t_i - \mu}{\sigma} \right) \right] \right\} \\ &\quad \times \prod_{S_2} \left\{ \frac{\exp \left\{ \exp \left(\frac{\tau_i^L - \mu}{\sigma} \right) \right\}}{\sigma} \exp \left[\left(\frac{t_i - \mu}{\sigma} \right) - \exp \left(\frac{t_i - \mu}{\sigma} \right) \right] \right\}. \end{aligned}$$

Correspondingly, the complete data log-likelihood function, using the truncation indicator v_i , is given by

$$\log L_c(\mathbf{t}; \theta) = -n \log \sigma + \sum_{i=1}^n \left[\left(\frac{t_i - \mu}{\sigma} \right) - \exp \left(\frac{t_i - \mu}{\sigma} \right) \right] + \sum_{i=1}^n (1 - v_i) \exp \left(\frac{\tau_i^L - \mu}{\sigma} \right). \quad (5.1)$$

In the E-step, the expectations of interest are $E_{\theta^{(k)}} \left[\frac{T_i - \mu}{\sigma} | T_i > y_i \right]$ and $E_{\theta^{(k)}} \left[\exp \left(\frac{T_i - \mu}{\sigma} \right) | T_i > y_i \right]$. To derive these conditional expectations, we first consider the conditional density of T_i , given $T_i > y_i$, given by

$$f_{T_i | (T_i > y_i)}(t_i) = \frac{\exp[\exp(\frac{y_i - \mu}{\sigma})]}{\sigma} \exp \left[\left(\frac{t_i - \mu}{\sigma} \right) - \exp \left(\frac{t_i - \mu}{\sigma} \right) \right], \quad t_i > y_i.$$

Based on the above conditional density function, the conditional mgf (moment generating function) of $\left(\frac{T_i - \mu}{\sigma} \right)$, given $T_i > y_i$, can be easily derived to be

$$M_{\left(\frac{T_i - \mu}{\sigma} \right)}(\theta) = \exp[\exp(\xi_i)] \Gamma(\theta + 1, e^{\xi_i}), \quad (5.2)$$

where $\xi_i = \left(\frac{y_i - \mu}{\sigma} \right)$ and $\Gamma(p, x) = \int_x^\infty u^{p-1} e^{-u} du$ is the upper incomplete gamma function.

Using the mgf in (5.2), the required expectations can be derived to be

$$E_{\theta^{(k)}} \left[\frac{T_i - \mu}{\sigma} | T_i > y_i \right] = \frac{\mu^{(k)} + \sigma^{(k)} E_{1i}^{(k)} - \mu}{\sigma}, \quad (5.3)$$

$$E_{\theta^{(k)}} \left[\exp \left(\frac{T_i - \mu}{\sigma} \right) | T_i > y_i \right] = e^{\left(\frac{\mu^{(k)} - \mu}{\sigma} \right)} \exp[\exp(\xi_i^{(k)})] \Gamma \left(\frac{\sigma^{(k)}}{\sigma} + 1, e^{\xi_i^{(k)}} \right), \quad (5.4)$$

where

$$E_{1i}^{(k)} = e^{\xi_i^{(k)}} \Psi(1) - \sum_{p=0}^{\infty} \{ \xi_i^{(k)} + \Psi(1) \} \frac{e^{(p+1)\xi_i^{(k)}}}{\Gamma(p+2)} + \sum_{p=0}^{\infty} \frac{e^{(p+1)\xi_i^{(k)}} \Psi(p+2)}{\Gamma(p+2)}.$$

Using (5.3) and (5.4), we obtain the function to be maximized as

$$\begin{aligned} Q(\theta, \theta^{(k)}) = & -n \log \sigma + \sum_{i=1}^n (1 - v_i) \exp \left(\frac{\tau_i^L - \mu}{\sigma} \right) + \sum_{\delta_i=1} \left\{ \left(\frac{t_i - \mu}{\sigma} \right) - \exp \left(\frac{t_i - \mu}{\sigma} \right) \right\} \\ & + \sum_{\delta_i=0} \left\{ \frac{\mu^{(k)} + \sigma^{(k)} E_{1i}^{(k)} - \mu}{\sigma} \right\} - \sum_{\delta_i=0} \left\{ e^{\left(\frac{\mu^{(k)} - \mu}{\sigma} \right)} M_{\left(\frac{T_i - \mu}{\sigma} \right)} \left(\frac{\sigma^{(k)}}{\sigma} \right) \right\}. \end{aligned} \quad (5.5)$$

Clearly, maximizing (5.5) poses a challenge as the function involved is a very complicated non-linear function of μ and σ , and the process used for this purpose is described next. In the M-step, here we make use of the EM gradient algorithm (Lange, 1995). In this algorithm, the function $Q(\theta, \theta^{(k)})$ is maximized by a one-step Newton-Raphson method, to get the updated estimate $\theta^{(k+1)}$. For the detailed expressions which are used to carry out this one-step NR method, one may refer to Balakrishnan and Mitra (2012).

We also use the NR method for comparative purpose. In our study, we employed the NR method by the default function “maxNR” of the R software. We observed in our empirical study that the EM and the NR methods yield quite close results in most cases.

5.2. Asymptotic variances and covariance of the MLEs

The elements of the complete information matrix $I_T(\theta)$ are given by

$$-E\left[\frac{\partial^2}{\partial \mu^2} \log L_c(\mathbf{t}; \theta)\right] = \frac{n}{\sigma^2},$$

$$\begin{aligned} -E\left[\frac{\partial^2}{\partial \sigma^2} \log L_c(\mathbf{t}; \theta)\right] = & -\frac{n}{\sigma^2} - \sum_{i=1}^n \frac{v_i}{\sigma^2} [2\{\Psi(1) - \Psi(2)\} - \{\Psi^2(2) + \Psi'(2)\}] \\ & - \sum_{i=1}^n \frac{(1-v_i)}{\sigma^2} \left[2\{\lambda_i + \Psi(1) - \lambda_i \exp(e^{\lambda_i}) + \sum_{p=0}^{\infty} \frac{e^{(p+1)\lambda_i} \Psi(p+2)}{\Gamma(p+2)}\} \right. \\ & - 2\{\Psi(2)(1 + e^{\lambda_i}) + \sum_{p=0}^{\infty} \frac{e^{(p+2)\lambda_i} \Psi(p+3)}{\Gamma(p+3)} - \lambda_i(\exp(e^{\lambda_i}) - 1 - e^{\lambda_i})\} \\ & - \{(\Psi'(2) + \Psi^2(2))(1 + e^{\lambda_i}) - (\lambda_i^2 + 2\lambda_i \Psi(2))(\exp(e^{\lambda_i}) - 1 - e^{\lambda_i}) \\ & + 2(\lambda_i + \Psi(2)) \sum_{p=0}^{\infty} \frac{e^{(p+2)\lambda_i} \Psi(p+3)}{\Gamma(p+3)} + \sum_{p=0}^{\infty} \frac{e^{(p+2)\lambda_i} \{\Psi'(p+3) - \Psi^2(p+3)\}}{\Gamma(p+3)}\} \left. \right] \\ & - \sum_{i=1}^n \frac{(1-v_i)}{\sigma^2} [2\lambda_i e^{\lambda_i} + \lambda_i^2 e^{\lambda_i}], \end{aligned}$$

$$-E\left[\frac{\partial^2}{\partial \sigma \partial \mu} \log L_c(\mathbf{t}; \theta)\right] = \frac{n\Psi(2)}{\sigma^2} + \sum_{i=1}^n \frac{(1-v_i)}{\sigma^2} \left[\Psi(2)e^{\lambda_i} + \sum_{p=0}^{\infty} \frac{e^{(p+2)\lambda_i} \Psi(p+3)}{\Gamma(p+3)} - \lambda_i(\exp(e^{\lambda_i}) - 1) \right],$$

where $\lambda_i = \frac{t_i - \mu}{\sigma}$.

The required conditional density for obtaining the missing information matrix can be derived as (see Ng et al., 2002)

$$f_{C_i|(C_i > y_i)}(c_i | C_i > y_i, \mu, \sigma) = \frac{\exp(e^{\xi_i})}{\sigma} \exp\left[\left(\frac{c_i - \mu}{\sigma}\right) - \exp\left(\frac{c_i - \mu}{\sigma}\right)\right], \quad c_i > y_i.$$

From this conditional density, we obtain the following expressions:

$$\frac{\partial^2}{\partial \mu^2} \log f_{C_i|(C_i > y_i)} = \frac{1}{\sigma^2} [e^{\xi_i} - e^{\beta_i}], \quad (5.6)$$

$$\frac{\partial^2}{\partial \sigma^2} \log f_{C_i|(C_i > y_i)} = \frac{1}{\sigma^2} [1 + 2\xi_i e^{\xi_i} + \xi_i^2 e^{\xi_i} + 2\beta_i - 2\beta_i e^{\beta_i} - \beta_i^2 e^{\beta_i}], \quad (5.7)$$

$$\frac{\partial^2}{\partial \mu \partial \sigma} \log f_{C_i|(C_i > y_i)} = \frac{1}{\sigma^2} [1 + e^{\xi_i} + \xi_i e^{\xi_i} - e^{\beta_i} - \beta_i e^{\beta_i}], \quad (5.8)$$

where $\beta_i = \frac{C_i - \mu}{\sigma}$. To obtain the expected values of these second derivatives, as before, we find the conditional moment generating function of $\beta_i = \frac{C_i - \mu}{\sigma}$, given $C_i > y_i$. The conditional mgf is given

by

$$\begin{aligned} M_{\left(\frac{C_i - \mu}{\sigma}\right)}(\theta) &= \exp[\exp(\xi_i)] \Gamma(\theta + 1, e^{\xi_i}) \\ &= \Gamma(\theta + 1) \left[e^{e^{\xi_i}} - \sum_{p=0}^{\infty} \frac{e^{(\theta+p+1)\xi_i}}{\Gamma(\theta + p + 2)} \right]. \end{aligned}$$

From this expression, the following expectations can be obtained:

$$E[\beta_i | C_i > y_i] = \xi_i + \Psi(1) - \xi_i e^{e^{\xi_i}} + \sum_{p=0}^{\infty} \frac{e^{(p+1)\xi_i} \Psi(p+2)}{\Gamma(p+2)},$$

$$E[\exp(\beta_i) | C_i > y_i] = 1 + e^{\xi_i},$$

$$E[\beta_i \exp(\beta_i) | C_i > y_i] = \Psi(2)(1 + e^{\xi_i}) + \sum_{p=0}^{\infty} \frac{e^{(p+2)\xi_i} \Psi(p+3)}{\Gamma(p+3)} - \xi_i (e^{e^{\xi_i}} - 1 - e^{\xi_i}),$$

$$\begin{aligned} E[\beta_i^2 \exp(\beta_i) | C_i > y_i] &= [\Psi^2(2) + \Psi'(2)](1 + e^{\xi_i}) - [\xi_i^2 + 2\xi_i \Psi(2)](e^{e^{\xi_i}} - 1 - e^{\xi_i}) \\ &\quad + 2[\xi_i + \Psi(2)] \sum_{p=0}^{\infty} \frac{e^{(p+2)\xi_i} \Psi(p+3)}{\Gamma(p+3)} + \sum_{p=0}^{\infty} \frac{e^{(p+2)\xi_i} [\Psi'(p+3) - \Psi^2(p+3)]}{\Gamma(p+3)}. \end{aligned}$$

Using Eqs. (5.6) - (5.8) and the expressions of the above expectations, the observed information matrix can be easily derived as described in Section 3.1. Finally, by inverting $I_Y(\hat{\theta})$, the asymptotic variance-covariance matrix of the MLEs can be obtained; see Balakrishnan and Mitra (2012) for details.

6. Inference for the gamma distribution

The density function of X , where X is a gamma random variable with scale parameter θ and shape parameter κ , is given by (see Johnson et al., 1994)

$$f_X(x) = \frac{x^{\kappa-1} \exp(-x/\theta)}{\theta^{\kappa} \Gamma(\kappa)}, \quad x > 0, \theta > 0, \kappa > 0.$$

The corresponding cumulative density function of X is

$$F_X(x) = \frac{\gamma(\kappa, x/\theta)}{\Gamma(\kappa)}, \quad x > 0, \theta > 0, \kappa > 0,$$

where $\gamma(p, x) = \int_0^x u^{p-1} e^{-u} du$ is the lower incomplete gamma function. The lifetime variable T is now assumed to follow a gamma distribution with the above density. As mentioned earlier, in this case, we choose to work with the lifetime data in the original scale itself.

6.1. Estimation of parameters

With $\mathbf{t}$ denoting the lifetime data and $\boldsymbol{\lambda} = (\kappa, \theta)$ denoting the parameter vector, and all other notation remaining the same as before, had there been no censoring, the complete data likelihood function would be

$$L_c(\mathbf{t}; \boldsymbol{\lambda}) = \prod_{i \in S_1} \left\{ \frac{t_i^{\kappa-1} e^{-t_i/\theta}}{\theta^\kappa \Gamma(\kappa)} \right\} \times \prod_{i \in S_2} \left\{ \frac{t_i^{\kappa-1} e^{-t_i/\theta}}{\theta^\kappa \Gamma(\kappa, \tau_i^L/\theta)} \right\}$$

where $\Gamma(p, x)$ is the upper incomplete gamma function, as before.

The log-likelihood function, using the truncation indicator v_i , then becomes

$$\log L_c(\mathbf{t}; \boldsymbol{\lambda}) = \sum_{i=1}^n [(\kappa - 1) \log t_i - \frac{t_i}{\theta} - \kappa \log \theta] - \sum_{i=1}^n [v_i \log \Gamma(\kappa) + (1 - v_i) \log \Gamma(\kappa, \tau_i^L/\theta)].$$

Let the current value of the parameter vector, at the r -th step, be $\lambda^{(r)}$. It can be clearly seen that the required expectations are $E_{1i}^{(r)} = E_{\lambda^{(r)}}[\log T_i | T_i > y_i]$ and $E_{2i}^{(r)} = E_{\lambda^{(r)}}[T_i | T_i > y_i]$. These expectations can then be obtained from the conditional distribution of T_i , given $T_i > y_i$. This conditional density is given by

$$f_{T_i | (T_i > y_i)}(t_i) = \frac{t_i^{\kappa-1} \exp(-t_i/\theta)}{\theta^\kappa \Gamma(\kappa, y_i/\theta)}, \quad t_i > y_i, \theta > 0, \kappa > 0. \quad (6.1)$$

Based on this conditional density, we can first of all easily find

$$E_{\lambda^{(r)}}[T_i | T_i > y_i] = \frac{\theta^{(r)} \Gamma(\kappa^{(r)} + 1, y_i/\theta^{(r)})}{\Gamma(\kappa^{(r)}, y_i/\theta^{(r)})}.$$

For the other expectation, it can be seen that

$$E_{\lambda^{(r)}}[\log T_i | T_i > y_i] = \log \theta^{(r)} + \frac{1}{\Gamma(\kappa^{(r)}, y_i/\theta^{(r)})} \int_{y_i/\theta^{(r)}}^{\infty} u_i^{\kappa^{(r)}-1} \log u_i e^{-u_i} du_i. \quad (6.2)$$

Following Geddes, Glasser, Moore and Scott (1990), we have

$$\int_{y_i/\theta^{(r)}}^{\infty} u_i^{\kappa^{(r)}-1} \log u_i e^{-u_i} du_i = \left[\frac{d}{da} \Gamma(a, y_i/\theta^{(r)}) \right]_{a=\kappa^{(r)}}. \quad (6.3)$$

Upon using (6.3) in Eq. (6.2) and using the derivative of upper incomplete gamma function as given in Balakrishnan and Mitra (2013), we get

$$\begin{aligned} E_{\lambda^{(r)}}[\log T_i | T_i > y_i] &= \log \theta^{(r)} + \frac{1}{\Gamma(\kappa^{(r)}, y_i/\theta^{(r)})} \left[\Psi(\kappa^{(r)}) \Gamma(\kappa^{(r)}) \left\{ 1 - e^{-y_i/\theta^{(r)}} \sum_{p=0}^{\infty} \frac{(y_i/\theta^{(r)})^{\kappa^{(r)}+p}}{\Gamma(\kappa^{(r)} + p + 1)} \right\} \right. \\ &\quad \left. - \Gamma(\kappa^{(r)}) e^{-y_i/\theta^{(r)}} \log(y_i/\theta^{(r)}) \sum_{p=0}^{\infty} \frac{(y_i/\theta^{(r)})^{\kappa^{(r)}+p}}{\Gamma(\kappa^{(r)} + p + 1)} \right. \\ &\quad \left. + \Gamma(\kappa^{(r)}) e^{-y_i/\theta^{(r)}} \sum_{p=0}^{\infty} \frac{(y_i/\theta^{(r)})^{\kappa^{(r)}+p} \Psi(\kappa^{(r)} + p + 1)}{\Gamma(\kappa^{(r)} + p + 1)} \right]. \end{aligned}$$

Using these two expectations, we then obtain

$$Q(\lambda, \lambda^{(r)}) = \left\{ \sum_{i:\delta_i=1} (\kappa-1) \log t_i + \sum_{i:\delta_i=0} (\kappa-1) E_{1i}^{(r)} \right\} - \left\{ \sum_{i:\delta_i=1} \frac{t_i}{\theta} + \sum_{i:\delta_i=0} \frac{E_{2i}^{(r)}}{\theta} \right\} - n\kappa \log \theta - \sum_{i=1}^n \left\{ v_i \log \Gamma(\kappa) + (1-v_i) \log \Gamma(\kappa, \tau_i^L/\theta) \right\}.$$

The objective now is to maximize $Q(\lambda, \lambda^{(r)})$ with respect to λ . The first-order derivatives of the Q -function with respect to θ and κ are, respectively,

$$\frac{\partial Q}{\partial \theta} = \sum_{i:\delta_i=1} \frac{t_i}{\theta^2} + \sum_{i:\delta_i=0} \frac{E_{2i}^{(r)}}{\theta^2} - \frac{n\kappa}{\theta} - \sum_{i=1}^n (1-v_i) \frac{\frac{\partial}{\partial \theta} \Gamma(\kappa, \tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)}, \quad (6.4)$$

$$\frac{\partial Q}{\partial \kappa} = \sum_{i:\delta_i=1} \log t_i + \sum_{i:\delta_i=0} E_{1i}^{(r)} - n \log \theta - \sum_{i=1}^n \left\{ v_i \Psi(\kappa) + (1-v_i) \frac{\frac{\partial}{\partial \kappa} \Gamma(\kappa, \tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)} \right\}. \quad (6.5)$$

Now, the upper incomplete gamma function can be easily differentiated with respect to θ by employing Leibnitz's rule, and we then obtain from (6.4) that

$$\theta = \frac{1}{n\kappa} \left[\sum_{i:\delta_i=1} t_i + \sum_{i:\delta_i=0} E_{2i}^{(r)} - \sum_{i=1}^n (1-v_i) \frac{(\tau_i^L)^\kappa e^{-\tau_i^L/\theta}}{\theta^{\kappa-1} \Gamma(\kappa, \tau_i^L/\theta)} \right]. \quad (6.6)$$

The RHS of Eq. (6.6) can be evaluated at the current parameter value $\lambda^{(r)}$ to obtain the updated parameter estimate $\theta^{(r+1)}$.

Upon expanding the incomplete gamma function as an infinite series, then differentiating and simplifying the resulting expression, (6.5) can be expressed as

$$\begin{aligned} \frac{\partial Q}{\partial \kappa} = & \sum_{i:\delta_i=1} \log t_i + \sum_{i:\delta_i=0} E_{1i}^{(r)} - n \log \theta - n \Psi(\kappa) \\ & - \sum_{i=1}^n (1-v_i) \left[\log(\tau_i^L/\theta) - \log(\tau_i^L/\theta) / \left\{ 1 - e^{-\tau_i^L/\theta} \sum_{p=0}^{\infty} \frac{(\tau_i^L/\theta)^{\kappa+p}}{\Gamma(\kappa+p+1)} \right\} \right] \\ & + e^{-\tau_i^L/\theta} \sum_{p=0}^{\infty} \frac{(\tau_i^L/\theta)^{\kappa+p} \Psi(\kappa+p+1)}{\Gamma(\kappa+p+1)} / \left\{ 1 - e^{-\tau_i^L/\theta} \sum_{p=0}^{\infty} \frac{(\tau_i^L/\theta)^{\kappa+p}}{\Gamma(\kappa+p+1)} \right\} \right]. \end{aligned}$$

Equating $\frac{\partial Q}{\partial \kappa}$ to zero, the equation can be solved numerically for κ to obtain the current estimate $\kappa^{(r+1)}$ by using $\theta^{(r+1)}$ for θ . The E-step and M-step are then repeated till convergence is achieved to the desired level of accuracy.

6.2. Asymptotic variances and covariance of the MLEs

The elements of the complete information matrix are given by

$$-E \left[\frac{\partial^2}{\partial \theta^2} \log L_c(\mathbf{t}; \lambda) \right] = \frac{n\kappa}{\theta^2} + \sum_{i=1}^n (1-v_i) \left[\frac{\tau_i^{L\kappa} e^{-\tau_i^L/\theta}}{\theta^{\kappa+2} \Gamma(\kappa, \tau_i^L/\theta)} \left\{ \frac{\tau_i^L}{\theta} - \kappa + 1 \right\} - \left\{ \frac{(\tau_i^L)^\kappa e^{-\tau_i^L/\theta}}{\Gamma(\kappa, \tau_i^L/\theta)} \right\}^2 \right],$$

$$\begin{aligned}
-E \left[\frac{\partial^2}{\partial \kappa^2} \log L_c(\mathbf{t}; \boldsymbol{\lambda}) \right] &= \sum_{i=1}^n \left[v_i \Psi'(\kappa) + (1 - v_i) \left\{ \frac{\frac{\partial^2}{\partial \kappa^2} \Gamma(\kappa, \tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)} - \left\{ \frac{\frac{\partial}{\partial \kappa} \Gamma(\kappa, \tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)} \right\}^2 \right\} \right], \\
-E \left[\frac{\partial^2}{\partial \kappa \partial \theta} \log L_c(\mathbf{t}; \boldsymbol{\lambda}) \right] &= \frac{n}{\theta} + \sum_{i=1}^n (1 - v_i) \left(\frac{\tau_i^{L\kappa}}{\theta^{\kappa+1}} \right) e^{-\tau_i^L/\theta} \left[\frac{\log(\tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)} - \frac{\frac{\partial}{\partial \kappa} \Gamma(\kappa, \tau_i^L/\theta)}{\Gamma(\kappa, \tau_i^L/\theta)^2} \right].
\end{aligned}$$

The conditional density, required for the derivation of the missing information matrix, can be derived to be

$$f_{C_i|(C_i > y_i)}(c_i | C_i > y_i, \theta, \kappa) = \frac{c_i^{\kappa-1} e^{-c_i/\theta}}{\theta^\kappa \Gamma(\kappa, y_i/\theta)}, \quad c_i > y_i, \theta > 0, \kappa > 0.$$

From this conditional density, we easily derive the following expressions:

$$-\frac{\partial^2}{\partial \theta^2} \log f_{C_i|(C_i > y_i)} = \frac{2c_i}{\theta^3} - \frac{\kappa}{\theta^2} + \frac{e^{-y_i/\theta} y_i^\kappa}{\Gamma(\kappa, y_i/\theta) \theta^{\kappa+2}} \left\{ \frac{y_i}{\theta} - \kappa - 1 \right\} - \left\{ \frac{e^{-y_i/\theta} (y_i^\kappa/\theta^{\kappa+1})}{\Gamma(\kappa, y_i/\theta)} \right\}^2, \quad (6.7)$$

$$-\frac{\partial^2}{\partial \kappa^2} \log f_{C_i|(C_i > y_i)} = \frac{\frac{\partial^2}{\partial \kappa^2} \Gamma(\kappa, y_i/\theta)}{\Gamma(\kappa, y_i/\theta)} - \left\{ \frac{\frac{\partial}{\partial \kappa} \Gamma(\kappa, y_i/\theta)}{\Gamma(\kappa, y_i/\theta)} \right\}^2, \quad (6.8)$$

$$-\frac{\partial^2}{\partial \kappa \partial \theta} \log f_{C_i|(C_i > y_i)} = \frac{1}{\theta} + \frac{e^{-y_i/\theta} y_i^\kappa \log(y_i/\theta)}{\Gamma(\kappa, y_i/\theta) \theta^{\kappa+1}} - \frac{e^{-y_i/\theta} (y_i^\kappa/\theta^{\kappa+1}) \frac{\partial}{\partial \kappa} \Gamma(\kappa, y_i/\theta)}{\Gamma(\kappa, y_i/\theta)^2}. \quad (6.9)$$

Clearly, the only required expectation here is $E[C_i | C_i > y_i]$ which is easily derived from the above conditional density to be

$$E[C_i | C_i > y_i] = \frac{\theta \Gamma(\kappa + 1, y_i/\theta)}{\Gamma(\kappa, y_i/\theta)}. \quad (6.10)$$

Using Eqs. (6.7) - (6.10), the missing information matrix can be obtained as described earlier in Section 3.1, using which the observed information matrix can be obtained without any difficulty. Finally, the asymptotic variance-covariance matrix of the MLEs can be obtained by inverting $I_Y(\hat{\boldsymbol{\lambda}})$.

The NR method is also employed for obtaining the MLEs for comparison purposes through the R-library function, `maxNR`. Using the observed information matrix, the asymptotic variances of the MLEs are estimated and the corresponding asymptotic confidence intervals are also obtained. It has been seen in the simulation study that the two methods perform quite closely.

7. Inference for the generalized gamma distribution

Let X be a generalized gamma random variable with scale parameter α and two shape parameters η and κ , respectively. Then, the density function of X is given by (see Johnson et al., 1994; Balakrishnan and Peng, 2006; Stacy, 1962)

$$f_X(x) = \frac{\eta}{\alpha \Gamma(\kappa)} \left(\frac{x}{\alpha} \right)^{\kappa\eta-1} \exp \left\{ - \left(\frac{x}{\alpha} \right)^\eta \right\}, \quad x > 0, \alpha > 0, \eta > 0, \kappa > 0.$$

The corresponding survival function of X is

$$S_X(x) = \frac{\Gamma(\kappa, (\frac{x}{\alpha})^\eta)}{\Gamma(\kappa)}, \quad x > 0, \alpha > 0, \eta > 0, \kappa > 0,$$

where $\Gamma(p, x)$ is the upper incomplete gamma function as introduced earlier.

The generalized gamma distribution contains exponential, gamma, Weibull and lognormal distributions, among others, as special cases. For $\eta = 1$ and $\kappa = 1$, it is exponential; for $\eta = 1$, it is gamma; and for $\kappa = 1$, it is the Weibull distribution. The lognormal distribution comes as a limiting case when $\eta \rightarrow 0$ and $\kappa \rightarrow \infty$. We now assume that the lifetime variable follows a generalized gamma distribution with the above density function. Here again, we work with the lifetime data in the original scale itself as it is most convenient to work with.

7.1. Estimation of parameters

With $\mathbf{t}$ denoting the lifetime data and $\boldsymbol{\theta} = (\alpha, \eta, \kappa)$ denoting the parameter vector, with all other notation remaining the same as before, had there been no censoring, the complete data likelihood function would be

$$L_c(\mathbf{t}; \boldsymbol{\theta}) = \prod_{i \in S_1} \left\{ \frac{\eta (\frac{t_i}{\alpha})^{\kappa\eta-1}}{\alpha \Gamma(\kappa)} \exp\{-(t_i/\alpha)^\eta\} \right\} \times \prod_{i \in S_2} \left\{ \frac{\eta (\frac{t_i}{\alpha})^{\kappa\eta-1}}{\alpha \Gamma(\kappa, (\frac{t_i}{\alpha})^\eta)} \exp\{-(t_i/\alpha)^\eta\} \right\}.$$

The log-likelihood function, using the truncation indicator v_i , then becomes

$$\begin{aligned} \log L_c(\mathbf{t}; \boldsymbol{\theta}) = & n(\log \eta - \log \alpha) + \sum_{i=1}^n [(\kappa\eta - 1) \log(t_i/\alpha) - (t_i/\alpha)^\eta] \\ & - \sum_{i=1}^n [v_i \log \Gamma(\kappa) + (1 - v_i) \log \Gamma(\kappa, (\frac{t_i}{\alpha})^\eta)]. \end{aligned}$$

We now need to obtain

$$Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{(r)}) = E_{\boldsymbol{\theta}^{(r)}}[\log L_c(\mathbf{t}; \boldsymbol{\theta}) | \mathbf{y}, \boldsymbol{\delta}]. \quad (7.1)$$

Clearly, the expectations needed in this case are

$$E_{\boldsymbol{\theta}^{(r)}}[\log(T_i/\alpha) | T_i > y_i] \quad \text{and} \quad E_{\boldsymbol{\theta}^{(r)}}[(T_i/\alpha)^\eta | T_i > y_i].$$

The conditional density of T_i , given $T_i > y_i$, required for these expectations, is given by

$$f_{T_i | (T_i > y_i)}(t_i) = \frac{\eta}{\alpha \Gamma(\kappa, (y_i/\alpha)^\eta)} \left(\frac{t_i}{\alpha}\right)^{\kappa\eta-1} \exp\left\{-\left(\frac{t_i}{\alpha}\right)^\eta\right\}, \quad t_i > y_i, \alpha > 0, \eta > 0, \kappa > 0. \quad (7.2)$$

Using this conditional density, we first of all readily obtain that

$$E_{\boldsymbol{\theta}^{(r)}}[(T_i/\alpha)^\eta | T_i > y_i] = \left\{ \frac{\alpha^{(r)}}{\alpha} \right\}^\eta \frac{\Gamma(\kappa^{(r)} + \eta/\eta^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})}{\Gamma(\kappa^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})}.$$

For the second required expectation, we see that

$$E_{\boldsymbol{\theta}^{(r)}}[\log(T_i/\alpha) | T_i > y_i] = E_{\boldsymbol{\theta}^{(r)}}[\log T_i | T_i > y_i] - \log \alpha = E_i^{(r)} - \log \alpha.$$

Now, note that

$$\begin{aligned} E_i^{(r)} &= E_{\theta^{(r)}}[\log T_i | T_i > y_i] \\ &= \log \alpha^{(r)} + \frac{1}{\eta^{(r)} \Gamma(\kappa^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})} \int_{(y_i/\alpha^{(r)})^{\eta^{(r)}}}^{\infty} u_i^{\kappa^{(r)}-1} \log u_i e^{-u_i} du_i. \end{aligned} \quad (7.3)$$

Using the result of Geddes et al. (1990), we have

$$\int_{(y_i/\alpha^{(r)})^{\eta^{(r)}}}^{\infty} u_i^{\kappa^{(r)}-1} \log u_i e^{-u_i} du_i = \left[\frac{d}{da} \Gamma(a, (y_i/\alpha^{(r)})^{\eta^{(r)}}) \right]_{a=\kappa^{(r)}}. \quad (7.4)$$

Upon substituting the expression of the integral in (7.4) into (7.3), we obtain

$$\begin{aligned} E_{\theta^{(r)}}[\log T_i | T_i > y_i] &= \log \alpha^{(r)} + \frac{1}{\eta^{(r)} \Gamma(\kappa^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})} \left[\Psi(\kappa^{(r)}) \Gamma(\kappa^{(r)}) \left\{ 1 - e^{-(y_i/\alpha^{(r)})^{\eta^{(r)}}} \right\} \right. \\ &\quad \times \sum_{p=0}^{\infty} \frac{(y_i/\alpha^{(r)})^{\eta^{(r)}(\kappa^{(r)}+p)}}{\Gamma(\kappa^{(r)}+p+1)} \left. \right\} - \Gamma(\kappa^{(r)}) e^{-(y_i/\alpha^{(r)})^{\eta^{(r)}}} \left\{ \log \{ (y_i/\alpha^{(r)})^{\eta^{(r)}} \} \right. \\ &\quad \times \sum_{p=0}^{\infty} \frac{(y_i/\alpha^{(r)})^{\eta^{(r)}(\kappa^{(r)}+p)}}{\Gamma(\kappa^{(r)}+p+1)} - \sum_{p=0}^{\infty} \frac{(y_i/\alpha^{(r)})^{\eta^{(r)}(\kappa^{(r)}+p)} \Psi(\kappa^{(r)}+p+1)}{\Gamma(\kappa^{(r)}+p+1)} \left. \right\} \Big]. \end{aligned}$$

Upon substituting the expressions for the above two expectations into (7.1), we obtain

$$\begin{aligned} Q(\theta, \theta^{(r)}) &= n(\log \eta - \log \alpha) + \left\{ \sum_{i:\delta_i=1} (\log t_i - \log \alpha) + \sum_{i:\delta_i=0} (E_i^{(r)} - \log \alpha) \right\} \\ &\quad - \left\{ \sum_{i:\delta_i=1} \left(\frac{t_i}{\alpha} \right)^{\eta} + \sum_{i:\delta_i=0} \left(\frac{\alpha^{(r)}}{\alpha} \right)^{\eta} \frac{\Gamma(\kappa^{(r)} + \eta/\eta^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})}{\Gamma(\kappa^{(r)}, (y_i/\alpha^{(r)})^{\eta^{(r)}})} \right\} \\ &\quad - \sum_{i=1}^n [v_i \log \Gamma(\kappa) + (1 - v_i) \log \Gamma(\kappa, (\tau_i^L/\alpha)^{\eta})]. \end{aligned} \quad (7.5)$$

Clearly, we can see from (7.5) that this maximization is quite involved. In this case, we use a slight variation of the EM-gradient algorithm. In the M-step, we use a two-stage process as follows. In the r -th step, first for a given value of the shape parameter $\kappa^{(r)}$, we perform a one-step Newton-Raphson on the Q -function to obtain the updated values $\alpha^{(r+1)}$ and $\eta^{(r+1)}$. Then, with these updated values, we perform another one-step Newton-Raphson on the Q -function to obtain the updated value $\kappa^{(r+1)}$. Then, with these updated values of all the parameters, we go back to the E-step and the algorithm is continued till convergence is achieved to the desired level of accuracy. The convergence criterion has been taken as the difference between the values of the Q -function in two successive steps is below a pre-set tolerance level.

In a Monte Carlo simulation study to be described in Section 8, it is observed that the numerical MLEs converge to the true parameter values, taking on an average 20 iterative steps. Here, a comment needs to be made on the convergence of the EM algorithm for the generalized gamma distribution under left truncation and right censoring. It has been pointed out to us by Drs. H. K. T. Ng and Z. Ye that for left truncated and right censored generalized gamma distribution, the convergence of the EM algorithm may be towards the local maxima. This point will be elaborated further in Section 9.4.

7.2. Asymptotic variances and covariances of the MLEs

The elements of the complete information matrix are as follows:

$$\begin{aligned}
 -E \left[\frac{\partial^2 \log L_c}{\partial \alpha^2} \right] &= -\frac{n\kappa\eta}{\alpha^2} + \frac{\eta(\eta+1)}{\alpha^2} \left[\sum_{i=1}^n \left\{ v_i \kappa + (1-v_i) \frac{\Gamma(\kappa+1, (\tau_i^L/\alpha)^\eta)}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right\} \right. \\
 &\quad + \sum_{i=1}^n (1-v_i) \left[\frac{e^{-(\tau_i^L/\alpha)^\eta} \frac{\eta}{\alpha^2} (\tau_i^L/\alpha)^\eta \kappa \{-\eta\kappa - 1 + \eta(\tau_i^L/\alpha)^\eta\}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right. \\
 &\quad \left. \left. - \left\{ \frac{\frac{\eta}{\alpha} (\tau_i^L/\alpha)^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right\}^2 \right] \right], \\
 -E \left[\frac{\partial^2 \log L_c}{\partial \eta^2} \right] &= \frac{n}{\eta^2} + \sum_{i=1}^n \left[v_i \frac{\kappa}{\eta^2} \{\Psi'(\kappa+1) + \Psi^2(\kappa+1)\} + (1-v_i) \frac{V_2(\kappa+1, \tau_i^L)}{\eta^2 \Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right] \\
 &\quad - \sum_{i=1}^n (1-v_i) \left[\frac{\{\log(\frac{\tau_i^L}{\alpha})\}^2 e^{-(\tau_i^L/\alpha)^\eta} (\frac{\tau_i^L}{\alpha})^\eta \kappa \{\kappa - (\frac{\tau_i^L}{\alpha})^\eta\}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right. \\
 &\quad \left. + \left\{ \frac{\log(\frac{\tau_i^L}{\alpha}) (\frac{\tau_i^L}{\alpha})^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right\}^2 \right], \\
 -E \left[\frac{\partial^2 \log L_c}{\partial \kappa^2} \right] &= \sum_{i=1}^n \left[v_i \Psi'(\kappa) + (1-v_i) \left\{ \frac{V_2(\kappa, \tau_i^L)}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} - \left\{ \frac{V_1(\kappa, \tau_i^L)}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right\}^2 \right\} \right], \\
 -E \left[\frac{\partial^2 \log L_c}{\partial \alpha \partial \eta} \right] &= \frac{n\kappa}{\alpha} - \frac{1}{\alpha} \sum_{i=1}^n \left[v_i (\kappa + \kappa \Psi(\kappa+1)) + (1-v_i) \left(\frac{\Gamma(\kappa+1, (\tau_i^L/\alpha)^\eta)}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right. \right. \\
 &\quad \left. \left. + \frac{V_1(\kappa+1, \tau_i^L)}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right) \right] - \sum_{i=1}^n \frac{(1-v_i)}{\alpha} \left[\frac{(\frac{\tau_i^L}{\alpha})^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \left\{ \eta \left(\frac{\tau_i^L}{\alpha} \right)^\eta \log \left(\frac{\tau_i^L}{\alpha} \right) \right. \right. \\
 &\quad \left. \left. - \eta \kappa \log \left(\frac{\tau_i^L}{\alpha} \right) - 1 \right\} - \frac{\frac{\eta}{\alpha} \log(\frac{\tau_i^L}{\alpha}) \{(\tau_i^L/\alpha)^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta}\}^2}{\{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)\}^2} \right], \\
 -E \left[\frac{\partial^2 \log L_c}{\partial \alpha \partial \kappa} \right] &= \frac{n\eta}{\kappa} + \sum_{i=1}^n (1-v_i) \left[\frac{\frac{\eta^2}{\alpha} (\frac{\tau_i^L}{\alpha})^\eta \kappa \log(\frac{\tau_i^L}{\alpha}) e^{-(\tau_i^L/\alpha)^\eta}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} - \frac{\frac{\eta}{\alpha} (\frac{\tau_i^L}{\alpha})^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta} V_1(\kappa, \tau_i^L)}{\{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)\}^2} \right], \\
 -E \left[\frac{\partial^2 \log L_c}{\partial \eta \partial \kappa} \right] &= -\sum_{i=1}^n \left[v_i \frac{\Psi(\kappa)}{\eta} + (1-v_i) \frac{V_1(\kappa, \tau_i^L)}{\eta \Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} \right] \\
 &\quad - \sum_{i=1}^n (1-v_i) \left[\frac{\eta \{\log(\frac{\tau_i^L}{\alpha})\}^2 (\frac{\tau_i^L}{\alpha})^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta}}{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)} - \frac{\log(\frac{\tau_i^L}{\alpha}) (\frac{\tau_i^L}{\alpha})^\eta \kappa e^{-(\tau_i^L/\alpha)^\eta} V_1(\kappa, \tau_i^L)}{\{\Gamma(\kappa, (\tau_i^L/\alpha)^\eta)\}^2} \right],
 \end{aligned}$$

where $\Psi(z) = \frac{d}{dz} \log \Gamma(z)$ and $\Psi'(z) = \frac{d^2}{dz^2} \log \Gamma(z)$ are the digamma and trigamma functions, respectively, and the functions $V_1(\cdot, \cdot)$ and $V_2(\cdot, \cdot)$ are given by

$$V_1(a, b) = \frac{\partial}{\partial u} \Gamma(u, (b/\alpha)^\eta) \Big|_{u=a},$$

$$V_2(a, b) = \frac{\partial^2}{\partial u^2} \Gamma(u, (b/\alpha)^\eta) \Big|_{u=a},$$

respectively.

The conditional density function required for the derivation of the missing information matrix can be derived to be

$$f_{C_i|(C_i > y_i)}(c_i | C_i > y_i, \alpha, \eta, \kappa) = \frac{\eta}{\alpha \Gamma(\kappa, (y_i/\alpha)^\eta)} \left(\frac{c_i}{\alpha} \right)^{\kappa\eta-1} \exp \left\{ - \left(\frac{c_i}{\alpha} \right)^\eta \right\}, \quad c_i > y_i, \alpha > 0, \eta > 0, \kappa > 0.$$

From this conditional density function, we obtain the following expressions:

$$\begin{aligned} -\frac{\partial^2}{\partial \alpha^2} \log f_{C_i|(C_i > y_i)} &= \frac{1}{\alpha^2} \left[-\frac{\eta(\eta\kappa+1)(y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} + \frac{\eta^2 (y_i/\alpha)^\eta \kappa + \eta e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \right. \\ &\quad \left. - \left\{ \frac{\eta (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \right\}^2 - \kappa\eta + \eta(\eta+1) \left(\frac{c_i}{\alpha} \right)^\eta \right], \\ -\frac{\partial^2}{\partial \eta^2} \log f_{C_i|(C_i > y_i)} &= \frac{1}{\eta^2} - \frac{\kappa \{\log(y_i/\alpha)\}^2 (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} + \frac{\{\log(y_i/\alpha)\}^2 (y_i/\alpha)^\eta \kappa + \eta e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \\ &\quad - \left\{ \frac{\log(y_i/\alpha) (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \right\}^2 + \left(\frac{c_i}{\alpha} \right)^2 \{\log(c_i/\alpha)\}^2, \\ -\frac{\partial^2}{\partial \kappa^2} \log f_{C_i|(C_i > y_i)} &= \frac{V_2(\kappa, y_i)}{\Gamma(\kappa, (y_i/\alpha)^\eta)} - \left\{ \frac{V_1(\kappa, y_i)}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \right\}^2, \\ -\frac{\partial^2}{\partial \alpha \partial \eta} \log f_{C_i|(C_i > y_i)} &= \frac{1}{\alpha} \left[\frac{(y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} + \frac{\log(y_i/\alpha) \eta \kappa (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} \right. \\ &\quad - \frac{\log(y_i/\alpha) (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta} \eta (y_i/\alpha)^\eta}{\Gamma(\kappa, (y_i/\alpha)^\eta)} + \frac{\eta \log(y_i/\alpha) \{(y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}\}^2}{\{\Gamma(\kappa, (y_i/\alpha)^\eta)\}^2} \\ &\quad \left. + \kappa - \eta \left(\frac{c_i}{\alpha} \right)^\eta \log(c_i/\alpha) - \left(\frac{c_i}{\alpha} \right)^\eta \right], \\ -\frac{\partial^2}{\partial \alpha \partial \kappa} \log f_{C_i|(C_i > y_i)} &= \frac{1}{\alpha} \left[\eta + \frac{\eta^2 \log(y_i/\alpha) (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} - \frac{\eta (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta} V_1(\kappa, y_i)}{\{\Gamma(\kappa, (y_i/\alpha)^\eta)\}^2} \right], \\ -\frac{\partial^2}{\partial \eta \partial \kappa} \log f_{C_i|(C_i > y_i)} &= -\frac{\eta \{\log(y_i/\alpha)\}^2 (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta}}{\Gamma(\kappa, (y_i/\alpha)^\eta)} + \frac{\log(y_i/\alpha) (y_i/\alpha)^\eta \kappa e^{-(y_i/\alpha)^\eta} V_1(\kappa, y_i)}{\{\Gamma(\kappa, (y_i/\alpha)^\eta)\}^2} \\ &\quad - \log(c_i/\alpha). \end{aligned}$$

Clearly, the expectations that are needed for the implementation of the missing information principle are

$$\begin{aligned} E[(C_i/\alpha)^\eta | C_i > y_i], E[\log(C_i/\alpha) | C_i > y_i], E[(C_i/\alpha)^\eta \log(C_i/\alpha) | C_i > y_i], \\ E[(C_i/\alpha)^\eta \{\log(C_i/\alpha)\}^2 | C_i > y_i]. \end{aligned}$$

Using the conditional density of C_i , given $C_i > y_i$, and the results of Geddes et al. (1990), along the same lines as in Section 3, we obtain

$$\begin{aligned} E[(C_i/\alpha)^\eta | C_i > y_i] &= \frac{\Gamma(\kappa + 1, (y_i/\alpha)^\eta)}{\Gamma(\kappa, (y_i/\alpha)^\eta)}, \\ E[\log(C_i/\alpha) | C_i > y_i] &= \frac{V_1(\kappa, y_i)}{\eta \Gamma(\kappa, (y_i/\alpha)^\eta)}, \\ E[(C_i/\alpha)^\eta \log(C_i/\alpha) | C_i > y_i] &= \frac{V_1(\kappa + 1, y_i)}{\eta \Gamma(\kappa, (y_i/\alpha)^\eta)}, \\ E[(C_i/\alpha)^\eta \{\log(C_i/\alpha)\}^2 | C_i > y_i] &= \frac{V_2(\kappa + 1, y_i)}{\eta^2 \Gamma(\kappa, (y_i/\alpha)^\eta)}. \end{aligned}$$

Using these expressions, the observed information matrix can be derived and the asymptotic variance-covariance matrix of the MLEs can then be obtained by inverting the observed information matrix. However, we have observed that the numerical values of the estimated variances of the MLEs obtained from this method come out to be quite large, for this distribution. This issue is further discussed in the following section.

8. Simulation results

All simulations were performed by using the R software. First, a certain percentage for the truncation was fixed. With this fixed percentage of truncation, the installation years were sampled by unequal probability with-replacement sampling from an arbitrary set of years. For our simulation study, we fixed 1980 and 2008 as the years of truncation and censoring, respectively. The entire set of installation years was divided into two parts: (1960 – 1979) and (1980 – 1995). Unequal probabilities were allocated over the different years as follows: for the period 1960 – 1979, a probability of 0.15 was attached to each of the first five years, and the remaining probability was distributed equally over the remaining years of this period; for the period 1980 – 1995, a probability of 0.1 was attached to each of the first six years, and a probability of 0.04 was attached to each of the remaining years of this period. The lifetimes of the machines, in years, were simulated from the distribution under consideration. The lifetimes of the machines were then added to the installation years, to obtain the years of failure. Clearly, the year of failure of a machine determined whether the lifetime of the machine was censored or not. For the right censored units, the lifetime was taken as the minimum of the lifetime and the censoring time. Because of the left truncation of the data, if the year of failure is before 1980 for any particular unit, there is no information available on the lifetime of that unit. So, if the year of failure was obtained to be a year before 1980 for any unit, that observation was discarded and a new installation year and lifetime were then simulated. This setup produced, along with the desired level of truncation, sufficiently many censored observations. All the simulation results were based on 1 000 Monte Carlo runs.

8.1. Lognormal distribution

The data are log-transformed and the analyses were performed on the logarithmic scale in this case. For the details on this simulation study, one may refer to Balakrishnan and Mitra (2014). The sample

sizes used in the simulation study are 75, 100 and 200. We have observed that Method 2 sometimes shows a problem with convergence for smaller sample sizes (such as 50). Empirically, it has been observed that for samples of size at least 75, the method works well and never shows a problem with convergence. Method 1 and the NR method, however, never show any problem with convergence, even for smaller sample sizes. The truncation percentage was fixed at 30 and 60, respectively. Fixing the truncation percentage at two levels allows us to observe the behaviour of the model under heavy and moderate truncation amounts. The setup produced, along with the desired level of truncation, sufficiently many censored observations. Two different choices of the parameter vector $\theta = (\mu, \sigma)$ were used: (3.5, 0.5) and (3.0, 0.2). The method of moment estimates, i.e., the sample mean and sample standard deviation, were used as initial values in the optimization method required for the determination of the MLEs. The tolerance limit for convergence was fixed as 0.001. One can choose a smaller tolerance value which would result in a larger number of iterations but with a gradient vector even closer to zero.

It is observed in the simulation study that the two methods under the EM algorithm and the NR method yield very close results. Table 1 presents the bias and root mean square errors of the parameter estimates by different methods, obtained from the Monte Carlo simulations.

Table 1: Bias (B) and root mean square error (RMSE) for three different techniques used for parameter estimation in the model.

$n = 75$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$RMSE(\hat{\mu})$	$RMSE(\hat{\sigma})$
(3.5, 0.5)	30%	1	-0.012	-0.011	0.084	0.072
		2	-0.012	-0.011	0.084	0.072
		3	-0.010	-0.009	0.084	0.072
	60%	1	-0.012	-0.006	0.075	0.069
		2	-0.013	-0.006	0.076	0.069
		3	-0.012	-0.005	0.076	0.069
(3.0, 0.2)	30%	1	-0.006	-0.005	0.027	0.020
		2	-0.006	-0.005	0.027	0.020
		3	-0.006	-0.005	0.027	0.020
	60%	1	-0.009	-0.003	0.030	0.017
		2	-0.009	-0.002	0.030	0.017
		3	-0.009	-0.002	0.030	0.017

$n = 100$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$RMSE(\hat{\mu})$	$RMSE(\hat{\sigma})$
(3.5, 0.5)	30%	1	-0.014	-0.010	0.073	0.061
		2	-0.014	-0.010	0.073	0.061
		3	-0.012	-0.009	0.074	0.061
	60%	1	-0.009	-0.004	0.065	0.059
		2	-0.009	-0.004	0.065	0.059
		3	-0.008	-0.003	0.065	0.059
(3.0, 0.2)	30%	1	-0.006	-0.003	0.025	0.014
		2	-0.006	-0.004	0.025	0.017
		3	-0.006	-0.004	0.025	0.017
	60%	1	-0.009	-0.002	0.027	0.017
		2	-0.009	-0.002	0.027	0.017
		3	-0.009	-0.002	0.027	0.017

$n = 200$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$RMSE(\hat{\mu})$	$RMSE(\hat{\sigma})$
(3.5, 0.5)	30%	1	-0.020	-0.011	0.056	0.044
		2	-0.020	-0.012	0.056	0.044
		3	-0.018	-0.010	0.056	0.044
	60%	1	-0.013	-0.004	0.048	0.041
		2	-0.013	-0.005	0.049	0.041
		3	-0.012	-0.003	0.048	0.041
(3.0, 0.2)	30%	1	-0.007	-0.003	0.017	0.010
		2	-0.007	-0.003	0.017	0.010
		3	-0.007	-0.003	0.017	0.010
	60%	1	-0.009	-0.001	0.020	0.010
		2	-0.009	-0.001	0.020	0.010
		3	-0.009	-0.001	0.020	0.010

Note: Methods 1 and 2 stand for the two EM algorithm techniques, while Method 3 stands for the NR method.

It can be seen from Table 1 that the three methods give quite close results.

Table 2 gives the coverage probabilities of the confidence intervals for μ corresponding to Methods 1 and 2, the observed information matrix and the parametric bootstrap method, for different percentages of truncation and different nominal confidence levels.

Table 2: Coverage probabilities for different confidence intervals for μ .

<i>n</i> =75						
μ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
3.5	30%	90%	0.899	0.900	0.901	0.904
		95%	0.939	0.939	0.940	0.942
3.0	60%	90%	0.897	0.896	0.897	0.903
		95%	0.941	0.942	0.941	0.947
	30%	90%	0.889	0.888	0.889	0.902
		95%	0.935	0.935	0.935	0.943
	60%	90%	0.875	0.874	0.874	0.897
		95%	0.935	0.934	0.934	0.953
<i>n</i> =100						
μ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
3.5	30%	90%	0.887	0.886	0.891	0.880
		95%	0.941	0.940	0.946	0.944
3.0	60%	90%	0.901	0.899	0.901	0.895
		95%	0.946	0.945	0.946	0.943
	30%	90%	0.854	0.853	0.855	0.884
		95%	0.925	0.925	0.925	0.929
	60%	90%	0.868	0.863	0.863	0.894
		95%	0.935	0.934	0.934	0.948

$n=200$						
μ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
3.5	30%	90%	0.848	0.848	0.851	0.871
		95%	0.905	0.904	0.910	0.925
	60%	90%	0.863	0.862	0.864	0.876
		95%	0.923	0.923	0.923	0.932
3.0	30%	90%	0.875	0.875	0.875	0.886
		95%	0.926	0.926	0.926	0.934
	60%	90%	0.849	0.846	0.846	0.907
		95%	0.909	0.906	0.906	0.955

Analogous to Table 2, Table 3 gives the coverage probabilities for different confidence intervals for σ .

Table 3: Coverage probabilities for different confidence intervals for σ .

$n=75$						
σ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
0.5	30%	90%	0.872	0.871	0.879	0.879
		95%	0.926	0.925	0.929	0.937
	60%	90%	0.881	0.881	0.881	0.877
		95%	0.931	0.930	0.931	0.928
0.2	30%	90%	0.860	0.859	0.859	0.860
		95%	0.907	0.903	0.905	0.918
	60%	90%	0.894	0.893	0.893	0.893
		95%	0.943	0.942	0.943	0.937

$n=100$						
σ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
0.5	30%	90%	0.876	0.875	0.881	0.882
		95%	0.925	0.923	0.927	0.926
	60%	90%	0.899	0.898	0.900	0.881
		95%	0.939	0.939	0.940	0.938
0.2	30%	90%	0.902	0.897	0.901	0.901
		95%	0.956	0.951	0.951	0.955
	60%	90%	0.873	0.871	0.872	0.873
		95%	0.940	0.939	0.939	0.931

$n=200$						
σ	Truncation	Nominal CL	Coverage Probability			
			Method 1	Method 2	Observed	Bootstrap
0.5	30%	90%	0.875	0.874	0.881	0.879
		95%	0.921	0.920	0.926	0.937
	60%	90%	0.886	0.886	0.888	0.878
		95%	0.943	0.942	0.946	0.939
0.2	30%	90%	0.866	0.863	0.864	0.869
		95%	0.925	0.918	0.923	0.931
	60%	90%	0.888	0.888	0.888	0.878
		95%	0.940	0.940	0.940	0.929

Clearly, from Tables 2 and 3, it can be observed that the coverage probabilities corresponding to Methods 1 and 2 (using the missing information principle) and the observed information matrix are very close, and are close to the nominal level in almost all the cases considered. For larger sample

sizes (such as 200), the coverage probabilities stabilize at a level slightly lower than the nominal level. The same feature is observed for the parametric bootstrap confidence intervals as well. In fact, in some cases, the parametric bootstrap confidence intervals provide higher coverage probabilities compared to other asymptotic confidence intervals. However, the computation of parametric bootstrap confidence intervals takes much larger time and CPU usage compared to the others.

8.2. Weibull distribution

For the details regarding this simulation study interested readers may refer to Balakrishnan and Mitra (2012). The sample sizes used in this study are 100, 200 and 300. The truncation percentages are fixed at 30 and 40. Two choices of the Weibull parameter vector (α, η) are made: (35, 3) and (40, 4). Thus, the corresponding values for the extreme value parameter vector $\theta = (\mu, \sigma)$ are (3.55, 0.33) and (3.69, 0.25), respectively.

It is known that when T follows an extreme value distribution with parameters μ and σ , then

$$E(T) = \mu - \gamma\sigma, \quad \text{Var}(T) = \frac{\pi^2}{6}\sigma^2,$$

where $\gamma = 0.5772$ (approximately) is Euler's constant; see Johnson et al. (1995). From these expressions, the method of moments estimates for μ and σ can be determined easily. For all the values presented in Tables 5 – 7, the method of moments estimates for μ and σ were used as initial values in the optimization method required for the determination of the MLEs. Our objectives here are to obtain the estimates of the parameters and to observe the comparative performance of the methods of estimation. The bias and mean square error of the parameter estimates are obtained for all different methods of estimation.

As in the lognormal case, here also the EM gradient algorithm depends on the choice of initial values and shows a problem of convergence when a good choice is not made. However, it does converge to the true parameter value quite accurately when the initial values of the parameters are chosen well. For example, in our empirical study, we used the method of moments estimates as the initial values, and the EM gradient algorithm works very well with this choice. Also, for small sample sizes, this algorithm again faces problems with convergence. Through a simulation study, we noted the number of times the method failed to converge, and these are reported in Table 4.

Table 4: The number of times the EM algorithm failed to converge, in 1 000 Monte Carlo runs, in the extreme value model.

(μ, σ)	Trunc.	$n=100$	$n=200$	$n=300$
(3.55, 0.33)	30%	0	1	0
	40%	1	0	0
(3.69, 0.25)	30%	8	0	0
	40%	10	1	0

It can be noted that, for $n = 100$, the method failed to converge relatively often when the extreme value distribution parameters are 3.69 and 0.25. We have observed that for these simulation settings, the essential sample size (the observations which are neither truncated nor censored) sometimes could be as low as 4. Thus, divergence can occur due to this very small essential sample size. For larger sample sizes, the essential sample size also increases, thus increasing the chances of convergence of the method. Also, we observed in the simulation study that in a few cases estimates

given by the EM algorithm were highly biased, and substitution of those biased estimates into the missing information principle resulted in negative variance estimates; these cases were discarded from the simulation results. It was also observed that this happened only for smaller sample sizes.

Some comments on the parametric bootstrap confidence intervals need to be made here. For samples of size 100, carrying out the necessary calculations for the parametric bootstrap confidence intervals does not seem to be possible since the EM gradient algorithm has problems with convergence for this sample size. For samples of size 200 (or 300), the calculations can be possibly carried out for parametric bootstrap confidence intervals. However, we observed the time required for this computation to be unduly long, which prohibited us from evaluating the performance of this method. We plan to carry this out in the future with a parallel computing facility and hope to report those findings in a future paper.

The bias and mean square error of the parameter estimates corresponding to the EM algorithm and the NR methods are presented in Table 5. The tolerance limit to achieve convergence was set as 0.001. One can choose a smaller tolerance value which would result in a larger number of iterations but with a gradient vector even closer to zero.

Table 5: Bias (B) and mean square error (MSE) for the EM algorithm and the NR method.

$n = 100$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$MSE(\hat{\mu})$	$MSE(\hat{\sigma})$
(3.55, 0.33)	30%	EM	-0.014	-0.002	0.003	0.002
		NR	-0.012	-0.001	0.003	0.002
	40%	EM	-0.016	-0.004	0.003	0.002
		NR	-0.014	-0.003	0.003	0.002
(3.69, 0.25)	30%	EM	-0.014	-0.006	0.002	0.001
		NR	-0.011	-0.004	0.002	0.001
	40%	EM	-0.010	-0.004	0.002	0.001
		NR	-0.008	-0.002	0.002	0.001
$n = 200$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$MSE(\hat{\mu})$	$MSE(\hat{\sigma})$
(3.55, 0.33)	30%	EM	-0.015	-0.005	0.002	0.001
		NR	-0.013	-0.004	0.002	0.001
	40%	EM	-0.012	-0.003	0.001	0.001
		NR	-0.010	-0.001	0.001	0.001
(3.69, 0.25)	30%	EM	-0.014	-0.005	0.001	0.001
		NR	-0.011	-0.002	0.001	0.001
	40%	EM	-0.012	-0.005	0.001	0.001
		NR	-0.010	-0.003	0.001	0.001
$n = 300$						
(μ, σ)	Trunc.	Method	$B(\hat{\mu})$	$B(\hat{\sigma})$	$MSE(\hat{\mu})$	$MSE(\hat{\sigma})$
(3.55, 0.33)	30%	EM	-0.015	-0.004	0.001	0.001
		NR	-0.013	-0.002	0.001	0.001
	40%	EM	-0.013	-0.004	0.001	0.001
		NR	-0.012	-0.002	0.001	0.001
(3.69, 0.25)	30%	EM	-0.013	-0.004	0.001	0.000
		NR	-0.010	-0.002	0.001	0.000
	40%	EM	-0.011	-0.003	0.001	0.000
		NR	-0.009	-0.001	0.001	0.000

It can be observed from Table 5 that the bias and mean square error of the parameter estimates obtained from the two methods are quite close, with some occasional disagreement.

Table 6 gives the coverage probabilities of the asymptotic confidence intervals for μ corresponding to the EM algorithm and the NR method, for different nominal confidence levels and different simulation settings.

Table 6: Coverage probabilities for the two asymptotic confidence intervals for μ for different nominal confidence levels and different simulation settings.

n=100				
μ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
3.55	30%	90%	0.873*	0.882
		95%	0.933*	0.932*
	40%	90%	0.876*	0.874*
		95%	0.923*	0.927*
3.69	30%	90%	0.862*	0.868*
		95%	0.913*	0.920*
	40%	90%	0.879*	0.878*
		95%	0.934*	0.934*
n=200				
μ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
3.55	30%	90%	0.854*	0.860*
		95%	0.913*	0.918*
	40%	90%	0.867*	0.873*
		95%	0.936*	0.935*
3.69	30%	90%	0.838*	0.853*
		95%	0.913*	0.922*
	40%	90%	0.860*	0.862*
		95%	0.909*	0.924*
n=300				
μ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
3.55	30%	90%	0.839*	0.848*
		95%	0.907*	0.917*
	40%	90%	0.852*	0.860*
		95%	0.914*	0.916*
3.69	30%	90%	0.848*	0.865*
		95%	0.905*	0.913*
	40%	90%	0.857*	0.871*
		95%	0.927*	0.939

Note: The * values are significantly different from the nominal level of 95% confidence.

Analogous to Table 6, Table 7 presents the coverage probabilities for the two asymptotic confidence intervals for the parameter σ .

Table 7: Coverage probabilities for the two asymptotic confidence intervals for σ for different nominal confidence levels and different simulation settings.

$n=100$				
σ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
0.33	30%	90%	0.895	0.896
		95%	0.947	0.943
	40%	90%	0.875*	0.881*
		95%	0.918*	0.923*
0.25	30%	90%	0.898	0.906
		95%	0.933*	0.936*
	40%	90%	0.883	0.897
		95%	0.934*	0.940
$n=200$				
σ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
0.33	30%	90%	0.887	0.891
		95%	0.932*	0.938
	40%	90%	0.888	0.887
		95%	0.935*	0.943
0.25	30%	90%	0.890	0.910
		95%	0.937	0.948
	40%	90%	0.873*	0.883
		95%	0.928*	0.936*
$n=300$				
σ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
0.33	30%	90%	0.898	0.907
		95%	0.947	0.952
	40%	90%	0.905	0.908
		95%	0.947	0.950
0.25	30%	90%	0.879*	0.901
		95%	0.933*	0.943
	40%	90%	0.898	0.908
		95%	0.944	0.949

Note: The * values are significantly different from the nominal level of 95% confidence.

From Tables 6 and 7, it can be seen that the coverage probabilities for μ for the two methods are always close to each other and are lower than the nominal level. However, for σ , the coverage probabilities corresponding to the EM algorithm are always slightly higher than that of the NR method and are in general better for both methods. It can also be seen that for higher truncation percentages, the coverage probabilities for σ corresponding to the EM algorithm increase. This might be explained due to the fact that for higher truncation percentages, with the increase in the proportion of incomplete data, the confidence intervals for σ get wider. In general, however, the coverage probabilities for both μ and σ remain reasonably close to the nominal level, for both methods.

8.3. Gamma distribution

The sample sizes used in this study are 100, 200 and 300. The truncation percentages are fixed at 10 and 15. As the gamma distribution is right skewed, if the left truncation percentage is at a

higher level, then we would expect the estimates to be highly biased. The two choices of the gamma parameter vector (κ, θ) are made as (4, 8) and (5, 5). In this regard, it is useful to mention that Balakrishnan and Cohen (1991) observed that the standard errors of the gamma parameter estimates based on complete data were not reliable when the shape parameter value was less or equal to 2.5 as the corresponding information matrix becomes nearly singular. Here, in the case of left truncated and right censored data, empirically we observed the same as well. All the simulation results are based on 1 000 Monte Carlo runs.

It is known that, when T follows $\text{Gamma}(\kappa, \theta)$, we have (see Johnson et al., 1994)

$$E(T) = \theta \kappa, \quad \text{Var}(T) = \kappa \theta^2.$$

From these expressions, the moment estimates of θ and κ can be easily derived. However, these estimates are crude as the data are incomplete in our case. Using these crude moment estimates, we obtain the expected lifetimes of the censored observations. These expected lifetimes of the censored observations along with the observed lifetimes can be considered to form a pseudo-complete data set. Then, the moment estimates for the parameters are obtained from this pseudo-complete data, and are used as initial values in the optimization method required for the determination of the MLEs. This method is adopted in the iterative algorithm throughout. The bias and mean square error of the parameter estimates for the EM and NR methods are presented in Table 8.

Table 8: Bias (B) and mean square error (MSE) for the EM algorithm and the NR method.

$n = 100$						
(κ, θ)	Trunc.	Method	$B(\hat{\kappa})$	$B(\hat{\theta})$	$MSE(\hat{\kappa})$	$MSE(\hat{\theta})$
(4, 8)	10%	EM	0.394	-0.374	1.178	4.903
		NR	0.394	-0.373	1.180	4.910
	15%	EM	0.336	-0.313	1.112	4.560
		NR	0.337	-0.313	1.113	4.565
(5, 5)	10%	EM	0.375	-0.195	1.309	1.261
		NR	0.376	-0.196	1.311	1.262
	15%	EM	0.434	-0.279	1.353	1.160
		NR	0.435	-0.280	1.355	1.161
$n = 200$						
(κ, θ)	Trunc.	Method	$B(\hat{\kappa})$	$B(\hat{\theta})$	$MSE(\hat{\kappa})$	$MSE(\hat{\theta})$
(4, 8)	10%	EM	0.256	-0.397	0.539	2.478
		NR	0.256	-0.396	0.540	2.480
	15%	EM	0.254	-0.400	0.509	2.284
		NR	0.254	-0.399	0.510	2.287
(5, 5)	10%	EM	0.288	-0.229	0.662	0.653
		NR	0.289	-0.231	0.662	0.653
	15%	EM	0.286	-0.250	0.577	0.588
		NR	0.287	-0.251	0.578	0.588
$n = 300$						
(κ, θ)	Trunc.	Method	$B(\hat{\kappa})$	$B(\hat{\theta})$	$MSE(\hat{\kappa})$	$MSE(\hat{\theta})$
(4, 8)	10%	EM	0.217	-0.427	0.348	1.708
		NR	0.217	-0.426	0.348	1.709
	15%	EM	0.195	-0.368	0.336	1.580
		NR	0.195	-0.368	0.336	1.582
(5, 5)	10%	EM	0.204	-0.203	0.386	0.428
		NR	0.205	-0.204	0.387	0.428
	15%	EM	0.195	-0.194	0.362	0.396
		NR	0.196	-0.195	0.362	0.396

From Table 8, it is clearly seen that the two methods are in quite close agreement, in terms of bias and MSE of the parameter estimates.

Table 9 gives the coverage probabilities for θ corresponding to the two asymptotic confidence intervals based on the EM algorithm and the NR method. As the computational time required for the construction of parametric bootstrap confidence intervals is very high, it was skipped in the simulation study. But, for illustrative purposes, the parametric bootstrap confidence intervals are constructed in the next section for the numerical example considered.

Table 9: Coverage probabilities for the two asymptotic confidence intervals for θ for different nominal confidence levels.

$n=100$				
θ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
8	10%	90%	0.833*	0.833*
		95%	0.872*	0.872*
	15%	90%	0.837*	0.835*
		95%	0.872*	0.872*
5	10%	90%	0.830*	0.830*
		95%	0.883*	0.883*
	15%	90%	0.836*	0.836*
		95%	0.882*	0.882*
$n=200$				
θ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
8	10%	90%	0.834*	0.834*
		95%	0.881*	0.880*
	15%	90%	0.828*	0.828*
		95%	0.871*	0.871*
5	10%	90%	0.836*	0.836*
		95%	0.890*	0.890*
	15%	90%	0.839*	0.838*
		95%	0.905*	0.905*
$n=300$				
θ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
8	10%	90%	0.815*	0.815*
		95%	0.864*	0.864*
	15%	90%	0.841*	0.840*
		95%	0.896*	0.895*
5	10%	90%	0.840*	0.839*
		95%	0.901*	0.901*
	15%	90%	0.843*	0.843*
		95%	0.898*	0.898*

*Note: The * values are significantly different from the nominal level of 95% confidence.*

It can be seen from Table 9 that the coverage probabilities corresponding to the two methods are quite close, but they are consistently below the nominal level. Analogous to Table 9, Table 10 gives the coverage probabilities for the two confidence intervals for κ .

Table 10: Coverage probabilities for the two asymptotic confidence intervals for κ for different nominal confidence levels.

$n=100$				
κ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
4	10%	90%	0.908	0.909
		95%	0.951	0.951
	15%	90%	0.898	0.898
		95%	0.954	0.954
5	10%	90%	0.893	0.893
		95%	0.951	0.951
	15%	90%	0.890	0.890
		95%	0.951	0.951
$n=200$				
κ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
4	10%	90%	0.894	0.893
		95%	0.951	0.951
	15%	90%	0.886	0.885
		95%	0.95	0.95
5	10%	90%	0.884	0.884
		95%	0.932*	0.932*
	15%	90%	0.899	0.899
		95%	0.942	0.941
$n=300$				
κ	Truncation	Nominal CL	Coverage Probability	
			EM	NR
4	10%	90%	0.879*	0.879*
		95%	0.944	0.944
	15%	90%	0.880*	0.880*
		95%	0.948	0.948
5	10%	90%	0.889	0.889
		95%	0.939	0.939
	15%	90%	0.887	0.887
		95%	0.942	0.942

Note: The * values are significantly different from the nominal level of 95% confidence.

It can be seen from Table 10 that the coverage probabilities corresponding to the two methods are again quite close, and that they are very close to the nominal level as well.

8.4. Generalized gamma distribution

For the simulation study, the sample size used in this case is 200. The truncation percentage is fixed as 20, and the generalized gamma parameters used are $(\alpha, \eta, \kappa) = (15, 3, 5)$. This setup produced 50% of censored observations. The observation made by Balakrishnan and Cohen (1991) with regard to problems in the determination of the MLEs when the value of the shape parameter of a gamma distribution gets close to 2 is seen here as well. As the κ parameter of the generalized gamma is in fact the gamma shape parameter, we observe the same problem in the case of generalized gamma distribution also. Also, as the data are left truncated and right censored here, we chose the κ parameter to be 5 in our simulation study, in order to ensure that the asymptotic variances are stable

and reliable.

Also, we have observed that for higher censoring percentages, the asymptotic variances of the MLEs get overestimated and in these situations the corresponding coverage probabilities get close to unity.

For choosing the initial values of the parameters, we adopted a two-stage grid-search method. First, an interval around the true value of κ is divided into discrete numbers, and a number is chosen from that set at random, which is then used as the initial value for κ . Then, a two-dimensional grid is formed, of which the different points are the values of (α, η) , the intervals being taken around the true values of these parameters. Given the value of κ , the point in the two-dimensional grid at which the observed likelihood is maximized is then chosen as the initial value for (α, η) .

Table 11 presents the bias and mean square errors of the parameter estimates for the above simulation settings based on 1 000 Monte Carlo runs. The tolerance required for convergence of the algorithm was taken as 0.001. AT and AI in Table 11 denote the average tolerance and average number of iterations, respectively, over 1 000 Monte Carlo runs.

Table 11: Bias (B) and mean square error (MSE) for the MLEs based on EM algorithm when $n = 200$.

(α, η, κ)	$\hat{\alpha}$		$\hat{\eta}$		$\hat{\kappa}$		AT	AI
	Bias	MSE	Bias	MSE	Bias	MSE		
(15, 3, 5)	0.186	1.941	0.120	0.126	-0.015	0.414	0.0005	21.173

We can construct asymptotic confidence intervals for the parameters by using the empirical variances of the parameter estimates. The empirical variances are obtained from a Monte Carlo simulation with 1 000 runs, and then these variances are used for constructing the asymptotic confidence intervals for the parameters in another 1 000 Monte Carlo runs. We observed in such a simulation study that these empirical asymptotic confidence intervals also have coverage probabilities close to the nominal level, but not for the parameter κ . Table 12 presents the coverage probabilities for the asymptotic confidence intervals corresponding to the empirical method and the parametric bootstrap technique. Here, NCL denotes nominal confidence level.

Table 12: Coverage probabilities for asymptotic confidence intervals and parametric bootstrap confidence intervals when $n = 200$.

Parameter	NCL = 90%		NCL = 95%	
	Empirical	Bootstrap	Empirical	Bootstrap
$\alpha = 15$	0.915	0.905	0.970	0.962
$\eta = 3$	0.897	0.933	0.940	0.961
$\kappa = 5$	0.965	0.923	0.995	0.973

It can be seen from Table 11 that the bias and MSEs of the estimates are quite reasonably small. From Table 12, it can be observed that the parametric bootstrap confidence intervals have coverage probabilities close to the nominal level. The coverage probabilities of the empirical confidence intervals, though close to the nominal level for α and η , is not so for κ as they are too high. Thus, under this setup, with regard to confidence intervals of the parameters, we suggest the use of parametric bootstrap, though it is computationally quite demanding and that it may be difficult to use in case of small sample sizes as the estimation method may fail to converge in some instances and for large samples, the program takes immensely longer time to run.

9. Illustrative examples

9.1. Lognormal distribution

In this section, an example is given to illustrate the confidence intervals for the parameters developed in the preceding sections. The numerical example is based on a simulated data of size 100. The truncation percentage used in the data was 60, and the true value of the parameter was taken as $(\mu, \sigma) = (3.5, 0.5)$. The dataset used for this numerical example is the one given in Balakrishnan and Mitra (2014).

For this example, the initial values of the parameters, obtained by the method of moments, are (3.29, 0.38). The tolerance level was set as 0.000001. The three methods converged to the same final estimate (3.51, 0.51), correct up to two decimal places. Here, it needs to be mentioned that had we set the tolerance level at 0.001 as in the case of the simulation study, we would have terminated in fewer number of iterations (8 for Method 1, 12 for Method 2 and 5 for the NR method), with the same final estimate correct up to two decimal places.

Table 13 presents the asymptotic confidence intervals for μ and σ , corresponding to Methods 1 and 2 (using the missing information principle), the observed information matrix and the parametric bootstrap technique, for different nominal confidence levels.

Table 13: Different confidence intervals for μ and σ .

Parameter	Nominal CL	Method 1	Method 2	Observed	Bootstrap
$\mu = 3.5$	90%	(3.404, 3.622)	(3.404, 3.622)	(3.404, 3.622)	(3.413, 3.642)
	95%	(3.383, 3.643)	(3.383, 3.643)	(3.383, 3.643)	(3.391, 3.664)
$\sigma = 0.5$	90%	(0.412, 0.612)	(0.412, 0.612)	(0.412, 0.612)	(0.417, 0.623)
	95%	(0.392, 0.631)	(0.392, 0.631)	(0.392, 0.631)	(0.397, 0.643)

It can be seen from Table 13 that the confidence intervals corresponding to Methods 1 and 2 and the NR method are exactly the same, and this explains the close coverage probabilities for these methods in the last section. The confidence intervals corresponding to the parametric bootstrap are slightly wider than these three confidence intervals.

By employing the missing information principle, the complete information matrix for this data is obtained to be $\begin{pmatrix} 326.4489 & 107.6620 \\ 628.0912 & \end{pmatrix}$, and the observed information matrix to be $\begin{pmatrix} 229.4611 & -30.7460 \\ 274.4936 & \end{pmatrix}$. Thus, the ratio of the determinants of the observed information matrix to the complete information matrix is 0.3207, from which we could provide 67.93% as the proportion of information lost due to censoring. Alternatively, we can compute the corresponding variance-covariance matrices as $\begin{pmatrix} 0.0032 & -0.0006 \\ 0.0017 & \end{pmatrix}$ and $\begin{pmatrix} 0.0044 & 0.0005 \\ 0.0037 & \end{pmatrix}$ from which we can compute the trace-efficiency of the estimates based on censored data to complete data to be 60.49% and the determinant-efficiency to be 32.07%.

9.2. Weibull distribution

Here we use a sample of size 100, with truncation percentage 40. The true value of the parameter vector $\theta = (\mu, \sigma)$ is taken as (3.55, 0.33). The dataset used for this numerical example is the one in Balakrishnan and Mitra (2012). The moment estimates are taken as the initial values for μ and σ , which in this case are 3.340 and 0.271, respectively. The tolerance level was set here as 0.000001. The final estimate obtained by both methods is (3.537, 0.342), correct up to three decimal places. Here, it needs to be mentioned that had we set the tolerance level at 0.001, as in the case of the

simulation study, we would have terminated in 13 steps, with the estimate (3.536, 0.341), which would have been quite close to those obtained with the lower tolerance value.

Table 14 presents the asymptotic confidence intervals corresponding to the EM algorithm, the NR method, and the parametric bootstrap technique.

Table 14: Confidence intervals obtained by different methods.

Parameter	Nominal CL	EM	NR	Bootstrap
$\mu = 3.55$	90%	(3.458, 3.617)	(3.458, 3.617)	(3.465, 3.629)
	95%	(3.443, 3.632)	(3.443, 3.632)	(3.449, 3.644)
$\sigma = 0.33$	90%	(0.276, 0.407)	(0.275, 0.409)	(0.277, 0.411)
	95%	(0.264, 0.420)	(0.262, 0.422)	(0.264, 0.424)

It can be seen from Table 14 that while the confidence intervals for μ based on the EM and the NR methods are the same, the confidence intervals for σ based on the EM algorithm are narrower than those by the NR method. This explains the observed closeness of the coverage probabilities for μ for the two methods in Table 3, as well as the reason behind the slightly lower coverage probability for σ for the EM algorithm, observed in Table 4. The parametric bootstrap confidence intervals for μ are seen here to be slightly wider than the other two confidence intervals while for σ , it is comparable to the confidence intervals based on the other two methods. Thus, we would expect the parametric bootstrap method to have a higher coverage probability for μ .

By employing the missing information principle, the complete information matrix for this data is obtained to be $\begin{pmatrix} 855.7993 & 473.2701 \\ 1415.6032 & \end{pmatrix}$, and the observed information matrix to be $\begin{pmatrix} 427.8996 & -11.6697 \\ 630.8986 & \end{pmatrix}$. Thus, the ratio of the determinants of the observed information matrix to the complete information matrix is 0.2732, from which we could provide 72.68% as the proportion of information lost due to censoring. Alternatively, we can compute the corresponding variance-covariance matrices as $\begin{pmatrix} 0.0014 & -0.0005 \\ 0.0009 & \end{pmatrix}$ and $\begin{pmatrix} 0.0023 & 0.0000 \\ 0.0016 & \end{pmatrix}$ from which we can compute the trace-efficiency of the estimates based on censored data to complete data to be 58.97% and the determinant-efficiency to be 27.32%.

9.3. Gamma distribution

The sample size used for the illustration is 100, the truncation percentage is 15, and the true parameter value is $(\kappa, \theta) = (5, 5)$. The simulated data that is used for this illustration is presented Balakrishnan and Mitra (2013). The initial value, calculated by the method of moments explained in the preceding section, is found to be (3.293, 9.895). The tolerance level was set as 0.0001. The EM algorithm took 114 steps to give the final estimates of (κ, θ) to be (4.961, 4.897), while the NR method took 6 steps to converge to (4.962, 4.896). However, the time taken by the two methods in terms of CPU usage is almost the same. The asymptotic confidence intervals based on the EM algorithm, the NR method and the parametric bootstrap technique are all presented in Table 15.

Table 15: Confidence intervals obtained by different methods.

Parameter	NCL	EM	NR	Bootstrap
$\theta = 5$	90%	(3.212, 6.581)	(3.211, 6.580)	(3.453, 6.748)
	95%	(2.889, 6.904)	(2.888, 6.903)	(3.138, 7.064)
$\kappa = 5$	90%	(3.460, 6.461)	(3.461, 6.462)	(2.918, 6.286)
	95%	(3.173, 6.749)	(3.173, 6.750)	(2.596, 6.609)

From Table 15, we observe that confidence intervals based on the EM algorithm and the NR method are quite close. It is also observed that the parametric bootstrap confidence intervals are narrower compared to other asymptotic confidence intervals for θ , while they are slightly wider than the other two for the parameter κ .

By employing the missing information principle, the complete information matrix for this data is obtained to be $\begin{pmatrix} 20.6016 & 19.2816 \\ 20.0112 & \end{pmatrix}$, and the observed information is obtained to be $\begin{pmatrix} 17.7724 & 15.2849 \\ 14.0989 & \end{pmatrix}$. Thus, the ratio of the determinants of the observed information matrix to the complete information matrix is 0.4185, from which we could provide 58.15% as the proportion of information lost due to censoring. Alternatively, we can compute the corresponding variance-covariance matrices as $\begin{pmatrix} 0.4943 & -0.4763 \\ 0.5089 & \end{pmatrix}$ and $\begin{pmatrix} 0.8322 & -0.9022 \\ 1.0490 & \end{pmatrix}$ from which we can compute the trace-efficiency of the estimates based on censored data as compared to complete data to be 53.33% and the determinant-efficiency to be 41.85%.

9.4. Generalized gamma distribution

We consider a numerical example in this section to demonstrate the inferential methods developed in the preceding section for the generalized gamma model. For this purpose, we use a simulated sample of size 200, with truncation percentage 20 and true parameter value $(\alpha, \eta, \kappa) = (15, 3, 5)$. These data are presented in Table A.1 in the Appendix. The censoring percentage for this data turns out to be 49.5. The initial value, chosen as described in the preceding section, turns out to be (14.40, 3.0, 5.4). The tolerance level is set as 0.001. In 34 steps, the EM algorithm converges to the final estimate of (15.07, 3.18, 5.11).

As mentioned in Section 7.1, the numerical MLEs obtained by the EM algorithm for left truncated and right censored generalized gamma may converge to a local maxima. Drs. H. K. T. Ng and Z. Ye observed that a direct maximization method converged to (23.87, 6.27, 1.60) with log-likelihood value -296.31 for the same data, as compared to the above estimates obtained by the EM algorithm which gave a log-likelihood value of -297.87. Their observation was supported when they applied Stochastic EM (SEM) algorithm to the same data. The same has been observed for some other datasets also. For example, for another dataset with true parameter value (15, 3, 5) for (α, η, κ) , the EM algorithm converged to (17.52, 3.60, 4.20) with log-likelihood value -283.89, whereas the direct maximization method converged to (20.63, 4.44, 2.92) with log-likelihood value -283.81. For another dataset with true parameter values (25, 4, 2), the EM algorithm converged to (23.45, 3.65, 2.25) with log-likelihood -272.69, whereas the direct method converged to (24.79, 4.00, 1.97) with log-likelihood value -272.66. A reason for this issue may be the initial values given to the algorithm, as the algorithm that we have proposed is actually an application the EM gradient algorithm originally proposed by Lange (1995). Please note that in his paper, Lange (1995) did mention that the EM gradient algorithm is local in nature, and is dependent on the choice of initial values. Additional light shall be shed on this by the discussion of Drs. Ng and Ye. As they have used their SEM algorithm and has obtained the global maxima in this case, this is clearly an advantage of the SEM algorithm. We are grateful to them for pointing out this issue to us.

The empirical confidence intervals and the parametric bootstrap confidence intervals for the parameters are as presented in Table 16.

Table 16: Asymptotic confidence intervals and parametric confidence intervals for the parameters.

Parameter	Nominal CL	Empirical	Bootstrap
$\alpha = 15$	90%	(12.797, 17.337)	(12.829, 16.945)
	95%	(12.362, 17.772)	(12.435, 17.339)
$\eta = 3$	90%	(2.632, 3.730)	(2.515, 3.601)
	95%	(2.526, 3.835)	(2.411, 3.705)
$\kappa = 5$	90%	(4.057, 6.172)	(4.095, 6.176)
	95%	(3.854, 6.374)	(3.896, 6.376)

It can be clearly seen that the confidence intervals corresponding to the empirical method are slightly wider than the parametric bootstrap confidence intervals, from which the higher coverage probabilities for the empirical confidence intervals observed in Table 16 can be explained. In this case, the parametric bootstrap confidence interval seems to be a better choice as mentioned earlier, even though it is computationally much more demanding.

9.5. Prediction

We give an example based on the lognormal distribution to illustrate the method of prediction. Suppose that for a unit, Y_{ins} is 1961, i.e., the unit is left truncated; and it is right censored, with censoring year being 2008. The probability that this unit will be working till 2016 is estimated, as described in Section 3.3 using the estimates of μ and σ as in Section 9.1, to be 0.655. The standard error of this probability estimate, obtained by using Eq. (3.1) and the estimated variance-covariance matrix of the MLEs as $\begin{pmatrix} 0.0044 & 0.0005 \\ 0.0037 & \end{pmatrix}$, is 0.054. Consequently, an approximate 95% confidence interval for this probability is (0.549, 0.761). Similarly, for another unit for which the installation year is 1985 (i.e., not left truncated and also right censored), the probability that the unit will be working till 2016 is estimated to be 0.729, with a standard error of 0.035. Thus, as approximate 95% confidence interval for this probability is (0.660, 0.798). It may be of interest to note here that the second unit (installed in 1985) has a higher probability to work till 2016 than the first unit (installed in 1961), as one would naturally expect.

10. Model discrimination

As mentioned earlier, the generalized gamma family is a parsimonious family of distributions that includes some important lifetime models such as lognormal, exponential, Weibull and gamma as special cases. Thus, for a given data, it will be of interest to find out which of these special lifetime distributions provides the best fit. For this specific purpose, one can employ different techniques, such as the likelihood-based criteria and the information-based criteria. Here, we adopt the information-based criteria for this model discrimination problem.

10.1. Information criteria

Two of the most commonly used information-based criteria are Akaike's information criterion and the Bayesian information criterion.

Akaike's information criterion

Akaike's information criterion or AIC (Akaike, 1974) provides a way for model selection by evaluating an information measure based on the given data. It is given by

$$AIC = 2k - 2\log L,$$

where k is the number of parameters in the model and L is the maximized likelihood value (i.e., L evaluated at the determined values of the MLEs of the parameters). For a given data, when there are a number of models as potential candidates, the model with the lowest AIC is the best model to be fitted to that data, according to this criterion. Note that along with the maximized likelihood function, AIC contains a penalty term $2k$, which depends on the number of parameters in the model. Inclusion of this penalty term ensures that this criterion does not favor overfitting, i.e., for a given data, along with maximizing the likelihood function, one should also not fit a model with more parameters than what is essential.

The main advantages of AIC come from two different aspects. First of all, it is simple to apply, and easy to interpret. For a given data and a given set of models, one can calculate the AIC values, and then choose the model with minimum AIC to be the best model for the given data. Another important advantage of AIC is that it can be applied to models which are not nested; this comes mainly from the fact that the use of AIC is not linked to testing of hypotheses.

Bayesian information criterion

Another commonly used information-based method is the Bayesian information criterion or BIC (Schwarz, 1978) which is given by

$$BIC = k\log n - 2\log L,$$

where k and L are the number of model parameters and the likelihood function value evaluated at the determined values of the MLEs of the parameters as before, and n is the sample size. Observe that the penalty term in this case depends on both the number of parameters and the sample size.

We now make use of AIC and BIC for the model discrimination purpose. Through Monte Carlo simulations, we study the comparative performance of AIC and BIC for the models under consideration.

10.2. Simulation study

Our first objective is to discriminate between the three popular lifetime models - lognormal, Weibull and gamma. We maintain the same setup, for simulating left truncated and right censored samples from these distributions, as detailed earlier in Section 8. The procedure adopted here proceeds as follows. First of all, a left truncated and right censored sample from lognormal distribution is simulated and the parameters of the lognormal distribution are estimated based on that sample. Based on the same sample, the parameters of the Weibull and gamma distributions are also estimated. Then, by using these estimated parameters, the AIC and BIC for the respective models are calculated. As the sample is from the lognormal distribution, we would expect the AIC and BIC values corresponding to the lognormal model to be minimum among the three. This process is then repeated several

times, each time generating a new sample from the lognormal distribution, and the number of times the correct model (i.e., the lognormal) as well as the wrong model (i.e., the Weibull and gamma) is chosen is noted. It is expected that the lognormal model will be chosen as the appropriate model by AIC and BIC with higher probability than the other two competing models, of course!

A similar process is followed for samples from the Weibull and gamma distributions as well. Everytime a sample is chosen from one of these distributions, all three distributions are fitted to the data and the AIC and BIC values corresponding to these three models are then calculated. It is evident that this process gives rise to a 3×3 table, of which the diagonal elements represent the probability of choosing the correct model by AIC and BIC. The off-diagonal elements of this table represent the probability of choosing another model (from which the sample did not come).

For the lognormal distribution, the simulation parameters used are $(\mu, \sigma) = (3.5, 0.5)$ and truncation percentage is 60; for the Weibull distribution, the simulation parameters used are $(\alpha, \eta) = (35, 3)$ and truncation percentage is 40; for the gamma distribution, the simulation parameters used are $(\kappa, \theta) = (5, 5)$ and truncation percentage is 15. The sample sizes used for all the distributions are 100 and 300, and all the results are based on 500 Monte Carlo simulations. The results so obtained are presented in Table 17.

Table 17: Proportion of times the model is chosen by AIC and BIC when the data are from lognormal, Weibull and gamma models.

<i>n</i> =100				
Original model	Criteria	Fitted model		
		Lognormal	Weibull	Gamma
Lognormal	AIC	0.750	0.062	0.188
	BIC	0.750	0.062	0.188
Weibull	AIC	0.060	0.702	0.238
	BIC	0.060	0.702	0.238
Gamma	AIC	0.340	0.248	0.412
	BIC	0.340	0.248	0.412
<i>n</i> =300				
Original model	Criteria	Fitted model		
		Lognormal	Weibull	Gamma
Lognormal	AIC	0.796	0.008	0.196
	BIC	0.796	0.008	0.196
Weibull	AIC	0.000	0.864	0.136
	BIC	0.000	0.864	0.136
Gamma	AIC	0.180	0.118	0.702
	BIC	0.180	0.118	0.702

From Table 17, we observe that AIC and BIC perform without any difference for these models, which is to be expected since all the considered models have the same number of parameters and the same sample sizes. We observe that the diagonal elements, which estimate the probability of choosing the correct model, are always higher than the off-diagonal elements. This displays that AIC and BIC can successfully discriminate between these models under this setup. Also, we note that the probability of choosing the correct model increases with sample size. This means that, with increasing sample size, AIC and BIC become more accurate in determining the correct underlying distribution of the lifetimes based on left truncated and right censored data.

Next, we investigate the model discrimination problem from another perspective. We now simulate a left truncated and right censored data from the generalized gamma distribution, and fit all

four distributions to the data, i.e., lognormal, Weibull, gamma and generalized gamma. We calculate the AIC and BIC values for all the models, and based on those values, we determine which of these models is to be selected. We repeat this procedure 500 times, as before, and observe the proportion of times the different models are selected under this setting.

The setting used for the generalized gamma distribution in the simulation are: $(\alpha, \eta, \kappa) = (15, 3, 5)$, truncation percentage 20, and sample sizes 500 and 800. All the results are once again based on 500 Monte Carlo simulations. The results obtained from this simulation study are presented in Table 18.

Table 18: Proportion of times the model is chosen by AIC and BIC when the data are from generalized gamma distribution.

Original model: Generalized gamma	Criteria	Fitted model			
		Lognormal	Weibull	Gamma	Generalized gamma
$n = 500$	AIC	0.004	0.100	0.310	0.586
	BIC	0.004	0.238	0.672	0.086
$n = 800$	AIC	0.004	0.032	0.210	0.754
	BIC	0.004	0.102	0.526	0.368

From Table 18, we observe that the proportion of times the models are chosen by AIC always indicate that the generalized gamma model is chosen with higher probability than all other models; and this probability increases, as expected, when the sample size increases. However, according to BIC, in both cases, the model that is chosen most of the times is the gamma distribution. Even though the probability of selecting the generalized gamma distribution by BIC increases with sample size, it still remains less than the probability corresponding to the gamma distribution. This may be due to the fact that the penalty term in the BIC corresponding to the generalized gamma distribution (having three parameters) is generally more severe than that of the gamma distribution (having two parameters). Having said that, the simulation results also display that with an increase in the sample size, BIC is able to discriminate between the models more clearly. In general, based on the above results, we recommend the use of AIC for the model discrimination under this setup.

10.3. Illustrative examples

For the first illustrative example, we use the same datasets which were used in the numerical examples in Section 9 dealing with lognormal, Weibull and gamma distributions. For the lognormal distribution, the model parameters are $(\mu, \sigma) = (3.5, 0.5)$, the truncation percentage is 60, and the sample size is 100. Then, the lognormal, Weibull and gamma distributions are fitted to these data, and the AIC and BIC values corresponding to all these models are calculated. For the Weibull distribution, the model parameters are $(\alpha, \eta) = (35, 3)$, the truncation percentage is 40, and the sample size is 100. Similarly, for the gamma distribution, the model parameters are $(\kappa, \theta) = (5, 5)$, the truncation percentage is 15, and the sample size is 100. The same process is repeated for these two data sets as well. The AIC and BIC values so calculated are presented in Table 19, with the numbers in boldface being the minimum values.

Table 19: The AIC and BIC values for different models.

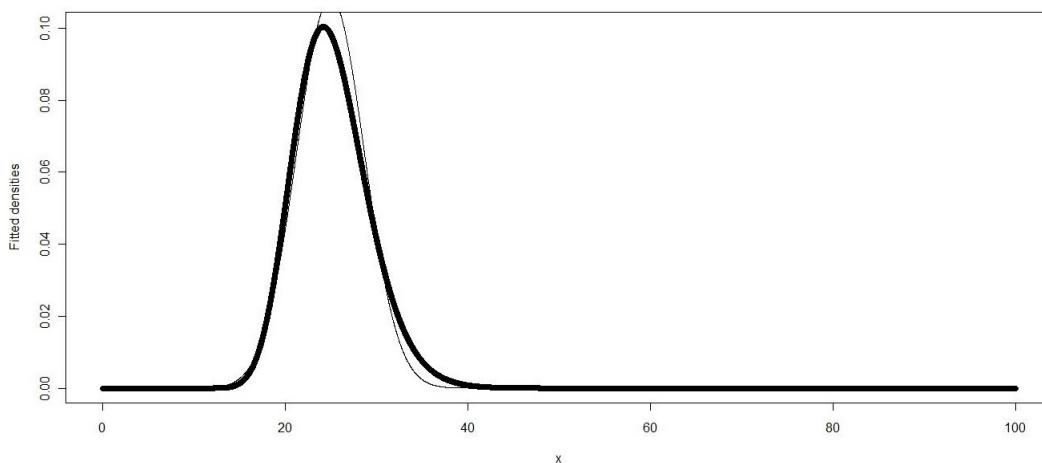
Original model	Criteria	Fitted model		
		Lognormal	Weibull	Gamma
Lognormal	AIC	423.2089	425.3809	423.6208
	BIC	428.4192	430.5913	428.8312
Weibull	AIC	425.2395	418.8227	421.5560
	BIC	430.4498	424.0330	426.7664
Gamma	AIC	453.2284	454.1567	451.9672
	BIC	458.4387	459.3670	457.1776

The model with the minimum AIC or BIC values is the most appropriate model, and it can be seen that the AIC and BIC values always agree with the correct model for all these three datasets.

To illustrate the model discrimination problem when the sample arises from the generalized gamma distribution, the model parameters used are $(\alpha, \eta, \kappa) = (15, 3, 5)$, truncation percentage is 20, and the sample size is 500. Table 20 presents the corresponding results, with the values in boldface being the minimum values once again. From Table 20, we do observe that the generalized gamma distribution is chosen by both methods as the best model to be fitted.

Table 20: The AIC and BIC values when the data is from the generalized gamma distribution.

Original data: Generalized gamma	Criteria	Fit			
		Lognormal	Weibull	Gamma	Generalized gamma
$n = 500$	AIC	1502.530	1496.578	1495.032	1489.220
	BIC	1510.959	1505.007	1503.461	1501.864

**Figure 1:** Fitted densities of the lognormal distribution (represented by the bold line) and the generalized gamma distribution (represented by the thin line).

Figures 1 to 3 show that when the data come from the generalized gamma distribution, the fitted densities for all the distributions are indeed quite close. In fact, for the above example, when the sample size is 500, the Kolmogorov-Smirnov statistic measuring the distances between the generalized gamma and each of lognormal, Weibull and gamma distributions turn out to be 0.038, 0.026 and 0.024, respectively. These show that the differences between the special cases and the generalized

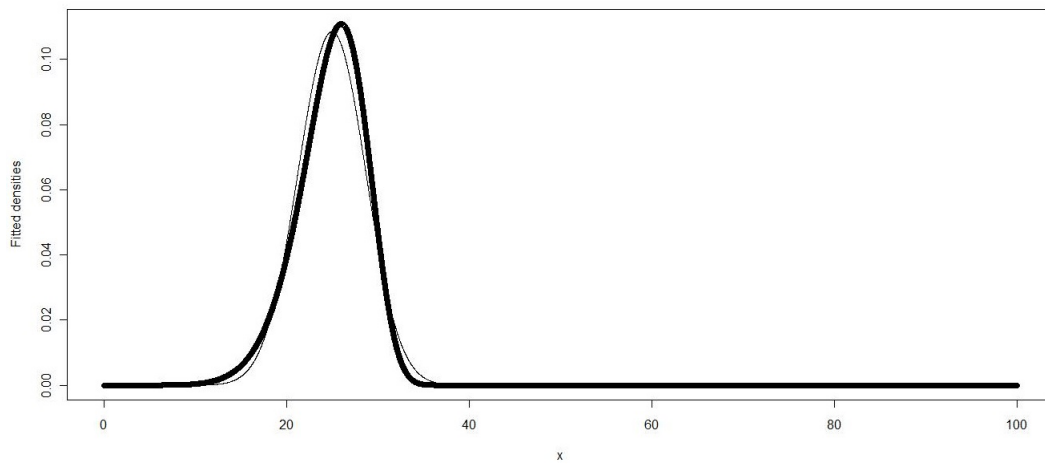


Figure 2: Fitted densities of the Weibull distribution (represented by the bold line) and the generalized gamma distribution (represented by the thin line)

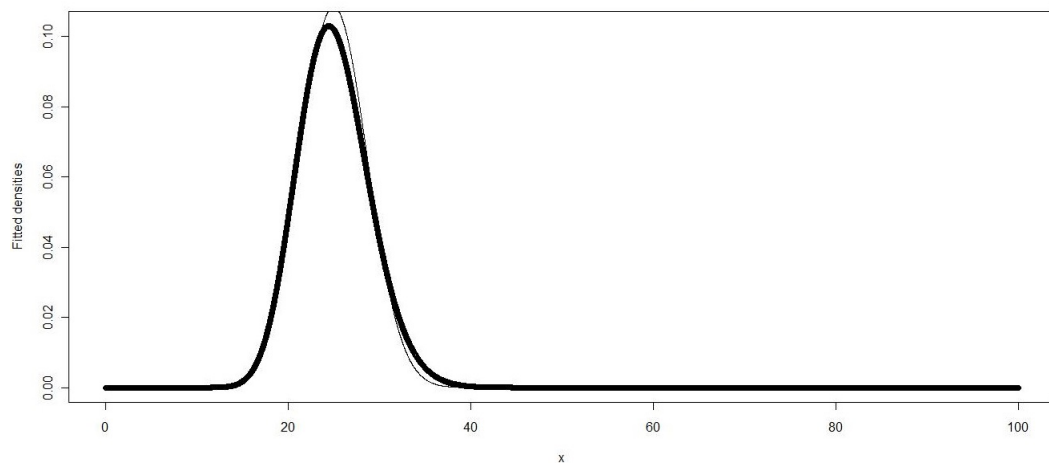


Figure 3: Fitted densities of the gamma distribution (represented by the bold line) and the generalized gamma distribution (represented by the thin line)

gamma distribution are indeed quite small, with the difference for the gamma distribution being the smallest.

11. Concluding remarks

In this paper, we develop the detailed steps of the EM algorithm for the lognormal, Weibull, gamma distributions based on left truncated and right censored data. The asymptotic variance-covariance

matrices of the MLEs are obtained by using the missing information principle. For all these models, the NR method is also used to obtain the MLEs, and then the two methods are compared through extensive Monte Carlo simulation studies. It is observed that the two methods always perform quite closely. The EM algorithm steps are also developed for the generalized gamma distribution based on left truncated and right censored data, and then a model discrimination problem under the parsimonious family of generalized gamma is carried out by using the information criteria. It is observed that though the AIC picks the correct model always, the BIC puts more importance on the number of parameters in the model and thus has a smaller probability of correct selection as compared to the AIC method. An application to prediction is also discussed, and some numerical examples are given.

An obvious extension of this work would be to include covariates into the models. The lifetimes of machines may depend on several factors like temperature, humidity, workload, etc. Inclusion of these factors into the model through an appropriate link function would surely enhance the applicability of this work. Another possible extension would be to consider interval censoring, instead of right censoring. The interval censoring may arise naturally from the sampling scheme, and it would be of interest to develop the EM algorithm steps for the left truncated and interval censored data. Yet another possible extension would be to consider semiparametric modelling in which one would model the data through an appropriate hazard function. We are working on these problems and hope to report these findings in some future papers.

Appendix

Table A.1 gives the dataset from the generalized gamma distribution which has been used as an illustrative example in Section 9.

Table A.1: A simulated dataset, with sample size 200, truncation percentage 20, and the true parameter value of (α, η, κ) as $(15, 3, 5)$, wherein * means not applicable.

Serial No.	Installation year	Truncation indicator	Truncation time	Failure year	Censoring indicator	Lifetime	Censoring time
1	1992	1	*	*	0	16	16
2	1991	1	*	*	0	17	17
3	1980	1	*	2004	1	24	*
4	1990	1	*	*	0	18	18
5	1985	1	*	*	0	23	23
6	1988	1	*	*	0	20	20
7	1982	1	*	2008	1	26	*
8	1993	1	*	*	0	15	15
9	1994	1	*	*	0	14	14
10	1987	1	*	*	0	21	21
11	1982	1	*	*	0	26	26
12	1986	1	*	*	0	22	22
13	1983	1	*	2006	1	23	*
14	1989	1	*	*	0	19	19
15	1984	1	*	*	0	24	24
16	1984	1	*	2006	1	22	*
17	1982	1	*	*	0	26	26

Serial No.	Installation year	Truncation indicator	Truncation time	Failure year	Censoring indicator	Lifetime	Censoring time
18	1982	1	*	2007	1	25	*
19	1992	1	*	*	0	16	16
20	1990	1	*	*	0	18	18
21	1983	1	*	*	0	25	25
22	1981	1	*	2006	1	25	*
23	1981	1	*	2002	1	21	21
24	1990	1	*	*	0	18	18
25	1981	1	*	2007	1	26	*
26	1986	1	*	*	0	22	22
27	1992	1	*	*	0	16	16
28	1988	1	*	*	0	20	20
29	1988	1	*	*	0	20	20
30	1985	1	*	*	0	23	23
31	1984	1	*	2006	1	22	22
32	1995	1	*	*	0	13	13
33	1991	1	*	*	0	17	17
34	1981	1	*	2006	1	25	*
35	1980	1	*	2007	1	27	*
36	1981	1	*	2002	1	21	*
37	1990	1	*	*	0	18	18
38	1992	1	*	*	0	16	16
39	1995	1	*	*	0	13	13
40	1992	1	*	2003	1	11	*
41	1983	1	*	2007	1	24	*
42	1995	1	*	*	0	13	13
43	1984	1	*	*	0	24	24
44	1981	1	*	*	0	27	27
45	1985	1	*	*	0	23	23
46	1991	1	*	*	0	17	17
47	1988	1	*	*	0	20	20
48	1981	1	*	2004	1	23	*
49	1984	1	*	*	0	24	24
50	1982	1	*	2003	1	21	*
51	1994	1	*	*	0	14	14
52	1985	1	*	*	0	23	23
53	1980	1	*	2005	1	25	*
54	1990	1	*	2007	1	17	*
55	1983	1	*	*	0	25	25
56	1983	1	*	*	0	25	25
57	1984	1	*	2007	1	23	*
58	1980	1	*	2005	1	25	*
59	1980	1	*	2004	1	24	*
60	1994	1	*	*	0	14	14
61	1984	1	*	2008	1	24	*
62	1993	1	*	*	0	15	15
63	1986	1	*	*	0	22	22
64	1982	1	*	2006	1	24	*
65	1982	1	*	2007	1	25	*
66	1982	1	*	*	0	26	26
67	1985	1	*	*	0	23	23
68	1984	1	*	*	0	24	24
69	1980	1	*	2003	1	23	*

Serial No.	Installation year	Truncation indicator	Truncation time	Failure year	Censoring indicator	Lifetime	Censoring time
70	1991	1	*	*	0	17	17
71	1993	1	*	*	0	15	15
72	1982	1	*	*	0	26	26
73	1993	1	*	*	0	15	15
74	1992	1	*	*	0	16	16
75	1984	1	*	2001	1	17	*
76	1989	1	*	*	0	19	19
77	1989	1	*	*	0	19	19
78	1986	1	*	2003	1	17	*
79	1988	1	*	*	0	20	20
80	1987	1	*	*	0	21	21
81	1981	1	*	2003	1	22	*
82	1992	1	*	*	0	16	16
83	1981	1	*	2005	1	24	*
84	1989	1	*	*	0	19	19
85	1984	1	*	*	0	24	24
86	1986	1	*	*	0	22	22
87	1994	1	*	*	0	14	14
88	1991	1	*	*	0	17	17
89	1987	1	*	2007	1	20	*
90	1980	1	*	2001	1	21	*
91	1988	1	*	*	0	20	20
92	1994	1	*	*	0	14	14
93	1987	1	*	*	0	21	21
94	1984	1	*	*	0	24	24
95	1989	1	*	2008	1	19	19
96	1981	1	*	*	0	27	27
97	1984	1	*	2006	1	22	*
98	1983	1	*	2007	1	24	*
99	1982	1	*	*	0	26	26
100	1983	1	*	2007	1	24	*
101	1984	1	*	*	0	24	24
102	1983	1	*	*	0	25	25
103	1991	1	*	*	0	17	17
104	1985	1	*	*	0	23	23
105	1983	1	*	2002	1	19	*
106	1989	1	*	*	0	19	19
107	1988	1	*	*	0	20	20
108	1993	1	*	*	0	15	15
109	1994	1	*	*	0	14	14
110	1982	1	*	2007	1	25	*
111	1981	1	*	2003	1	22	*
112	1993	1	*	*	0	15	15
113	1982	1	*	2004	1	22	*
114	1991	1	*	*	0	17	17
115	1981	1	*	2007	1	26	*
116	1983	1	*	2008	1	25	*
117	1987	1	*	*	0	21	21
118	1982	1	*	2004	1	22	*
119	1980	1	*	*	0	28	28
120	1995	1	*	*	0	13	13
121	1984	1	*	*	0	24	24

Serial No.	Installation year	Truncation indicator	Truncation time	Failure year	Censoring indicator	Lifetime	Censoring time
122	1985	1	*	*	0	23	23
123	1987	1	*	*	0	21	21
124	1980	1	*	2005	1	25	*
125	1992	1	*	*	0	16	16
126	1988	1	*	*	0	20	20
127	1982	1	*	*	0	26	26
128	1992	1	*	*	0	16	16
129	1982	1	*	2005	1	23	*
130	1981	1	*	2001	1	20	*
131	1984	1	*	2008	1	24	*
132	1981	1	*	1998	1	17	*
133	1980	1	*	2003	1	23	*
134	1988	1	*	2005	1	17	*
135	1991	1	*	*	0	17	17
136	1989	1	*	2006	1	17	*
137	1989	1	*	*	0	19	19
138	1985	1	*	2007	1	22	*
139	1986	1	*	2007	1	21	*
140	1980	1	*	2005	1	25	*
141	1988	1	*	*	0	20	20
142	1985	1	*	*	0	23	23
143	1984	1	*	2008	1	24	*
144	1994	1	*	*	0	14	14
145	1993	1	*	*	0	15	15
146	1984	1	*	2008	1	24	*
147	1982	1	*	2006	1	24	*
148	1982	1	*	2008	1	26	*
149	1983	1	*	*	0	25	25
150	1990	1	*	*	0	18	18
151	1995	1	*	*	0	13	13
152	1993	1	*	*	0	15	15
153	1982	1	*	2004	1	22	*
154	1984	1	*	2006	1	22	*
155	1987	1	*	*	0	21	21
156	1982	1	*	2008	1	26	*
157	1988	1	*	*	0	20	20
158	1989	1	*	*	0	19	19
159	1988	1	*	*	0	20	20
160	1980	1	*	2007	1	27	*
161	1961	0	19	1987	1	26	*
162	1960	0	20	1985	1	25	*
163	1968	0	12	1989	1	21	*
164	1977	0	3	1998	1	21	*
165	1965	0	15	1990	1	25	*
166	1964	0	16	1993	1	29	*
167	1961	0	19	1983	1	22	*
168	1977	0	3	2002	1	25	*
169	1961	0	19	1985	1	24	*
170	1961	0	19	1983	1	22	*
171	1971	0	9	1991	1	20	*
172	1961	0	19	1992	1	31	*
173	1966	0	14	1991	1	25	*

Serial No.	Installation year	Truncation indicator	Truncation time	Failure year	Censoring indicator	Lifetime	Censoring time
174	1976	0	4	2006	1	30	*
175	1979	0	1	2000	1	21	*
176	1960	0	20	1989	1	29	*
177	1960	0	20	1983	1	23	*
178	1961	0	19	1982	1	21	*
179	1962	0	18	1987	1	25	*
180	1962	0	18	1991	1	29	*
181	1973	0	7	1994	1	21	*
182	1961	0	19	1986	1	25	*
183	1979	0	1	2003	1	24	*
184	1960	0	20	1989	1	29	*
185	1977	0	3	2005	1	28	*
186	1963	0	17	1988	1	25	*
187	1964	0	16	1991	1	27	*
188	1965	0	15	1993	1	28	*
189	1960	0	20	1992	1	32	*
190	1977	0	3	1997	1	20	*
191	1962	0	18	1988	1	26	*
192	1963	0	17	1986	1	23	*
193	1974	0	6	1997	1	23	*
194	1979	0	1	2006	1	27	*
195	1962	0	18	1991	1	29	*
196	1970	0	10	1989	1	19	*
197	1974	0	6	2005	1	31	*
198	1960	0	20	1987	1	27	*
199	1977	0	3	2004	1	27	*
200	1964	0	16	1992	1	28	*

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